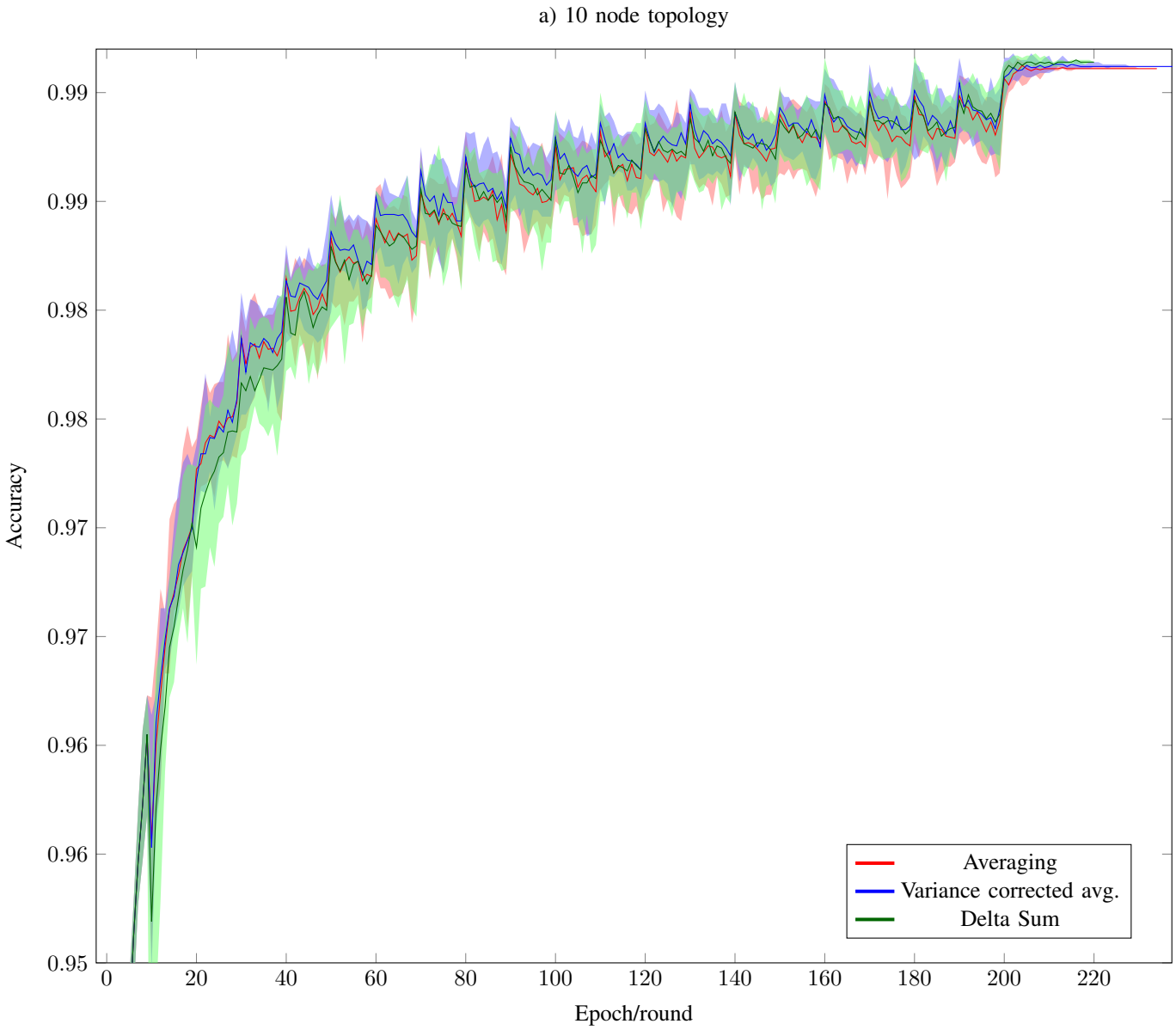



Fig. 1. Accuracy in a fully simulated topology for Delta Sum Learning and alternative strategies, with topology sizes ranging from 10 (a)) to 50 (c)). Comparing the different topology sizes, Delta Sum Learning results in a far lower accuracy loss than alternative strategies as topology size increases.

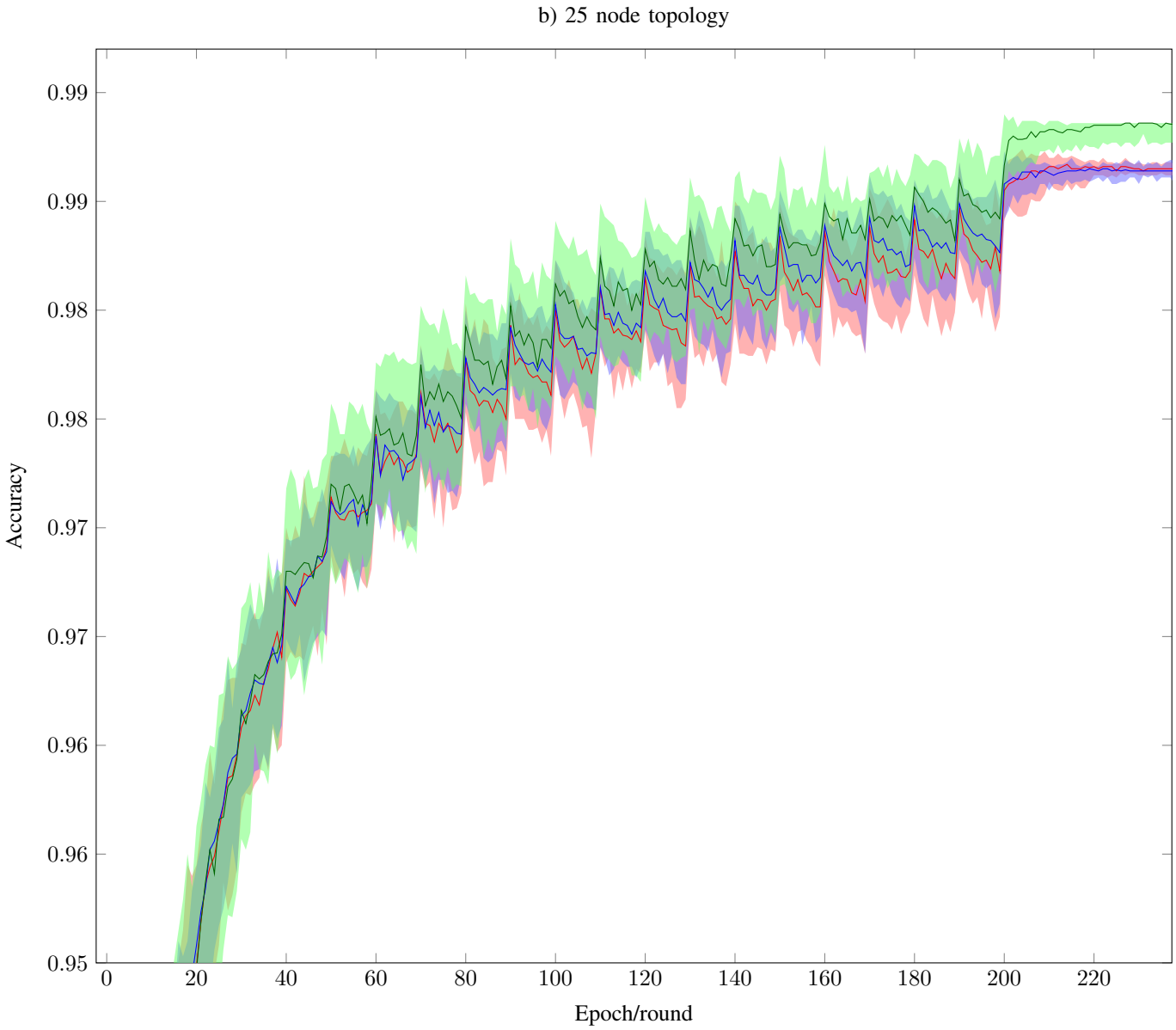



Fig. 2. Accuracy in a fully simulated topology for Delta Sum Learning and alternative strategies, with topology sizes ranging from 10 (a)) to 50 (c)). Comparing the different topology sizes, Delta Sum Learning results in a far lower accuracy loss than alternative strategies as topology size increases.

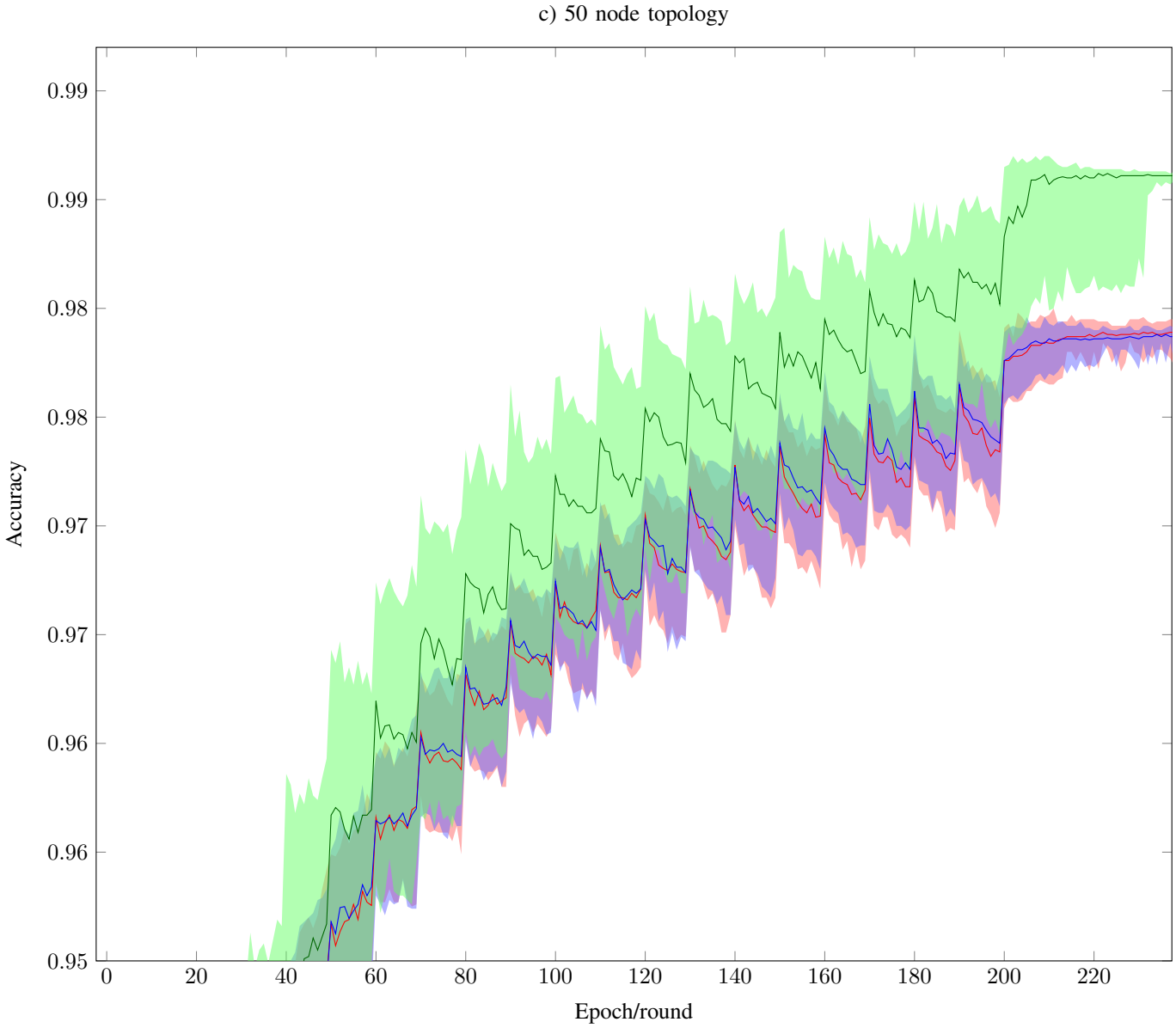



Fig. 3. Accuracy in a fully simulated topology for Delta Sum Learning and alternative strategies, with topology sizes ranging from 10 (a)) to 50 (c)). Comparing the different topology sizes, Delta Sum Learning results in a far lower accuracy loss than alternative strategies as topology size increases.

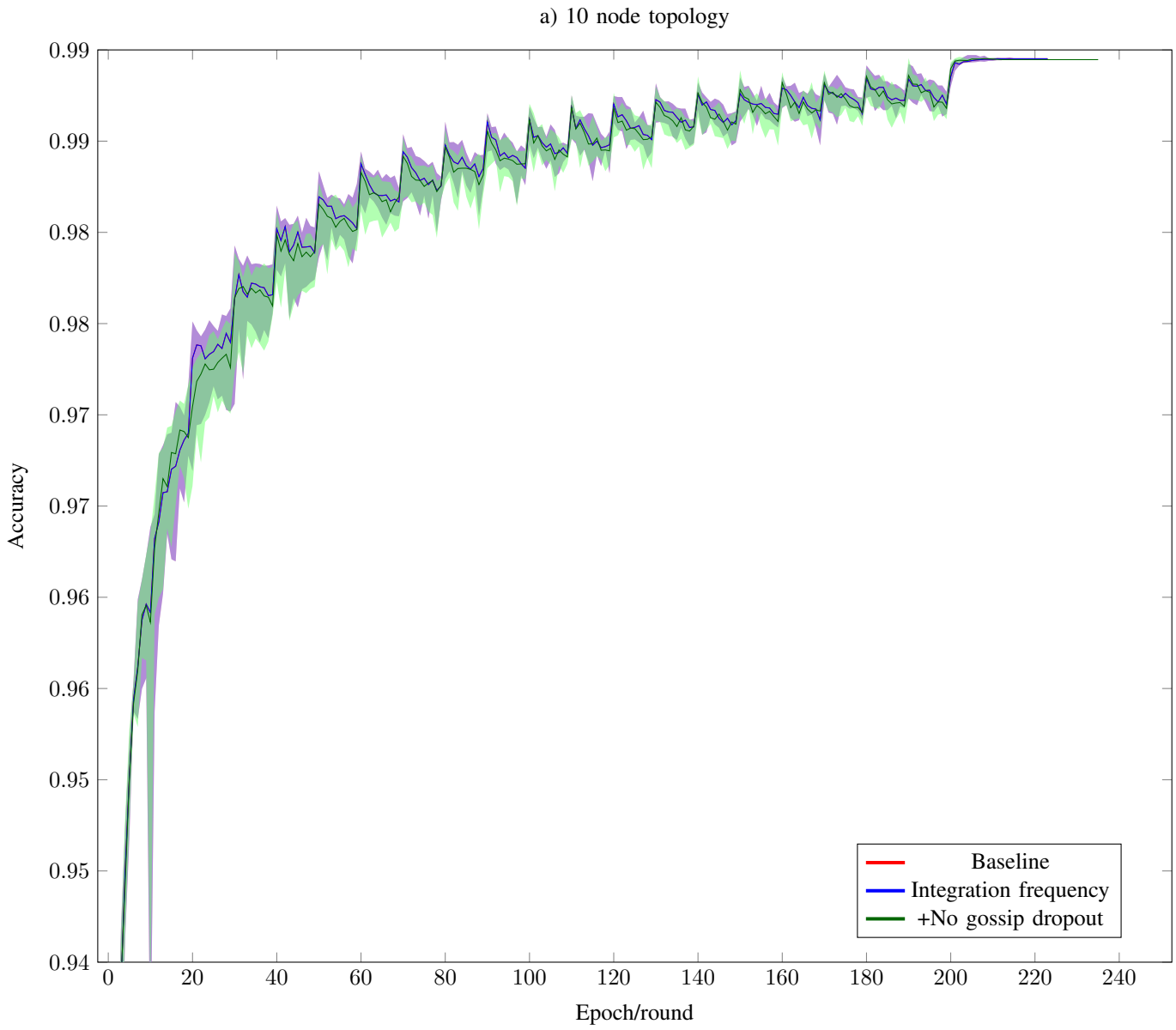



Fig. 4. Effect of increased update frequency $f \hat{=} \sqrt{n}$ and full message propagation, compared to baseline Delta Sum Learning (constant update frequency with default gossip cluster messaging behavior). While the update frequency by itself has little or no effect, the combination of both tweaks results in significantly better convergence.

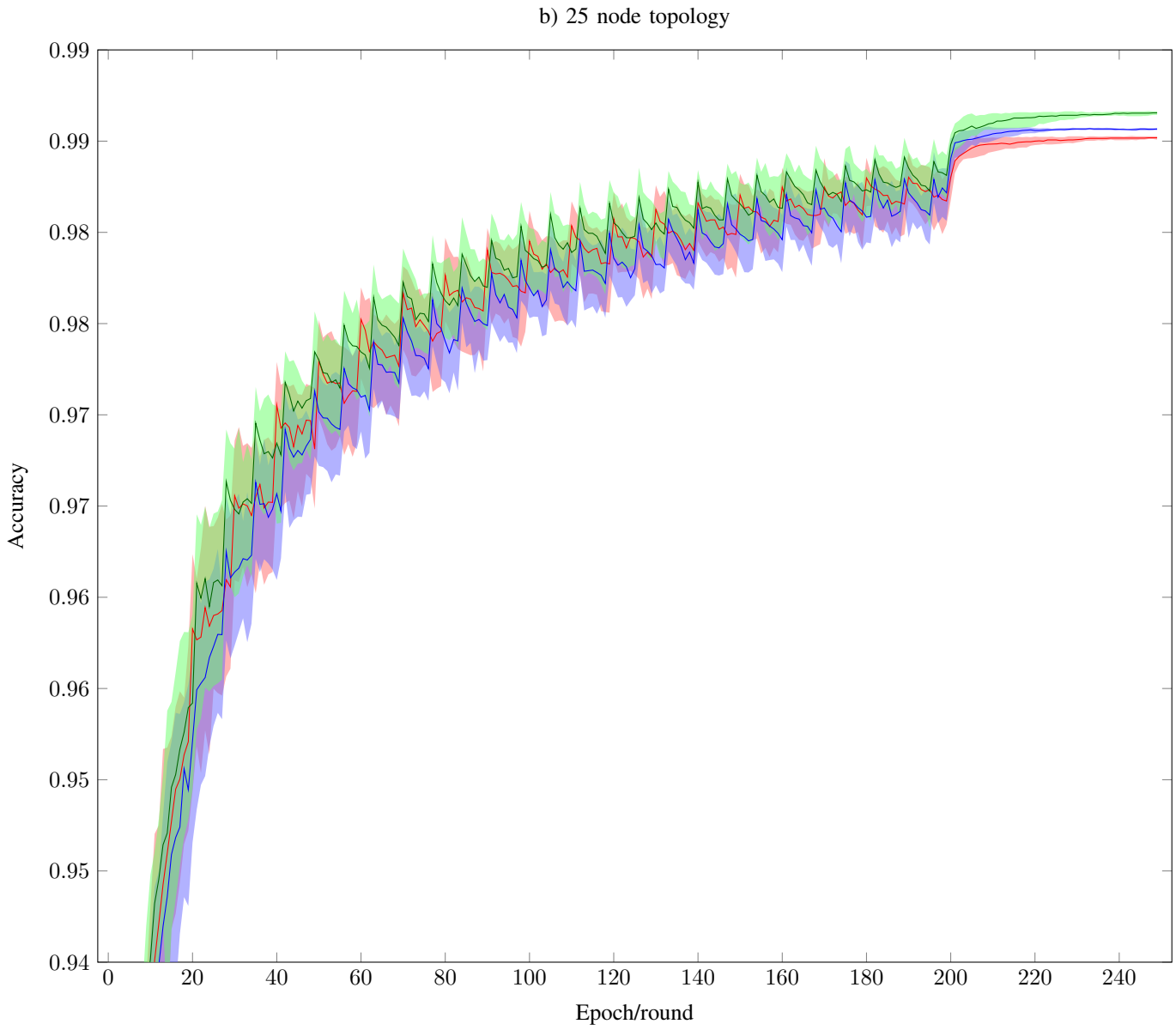



Fig. 5. Effect of increased update frequency $f \hat{=} \sqrt{n}$ and full message propagation, compared to baseline Delta Sum Learning (constant update frequency with default gossip cluster messaging behavior). While the update frequency by itself has little or no effect, the combination of both tweaks results in significantly better convergence.

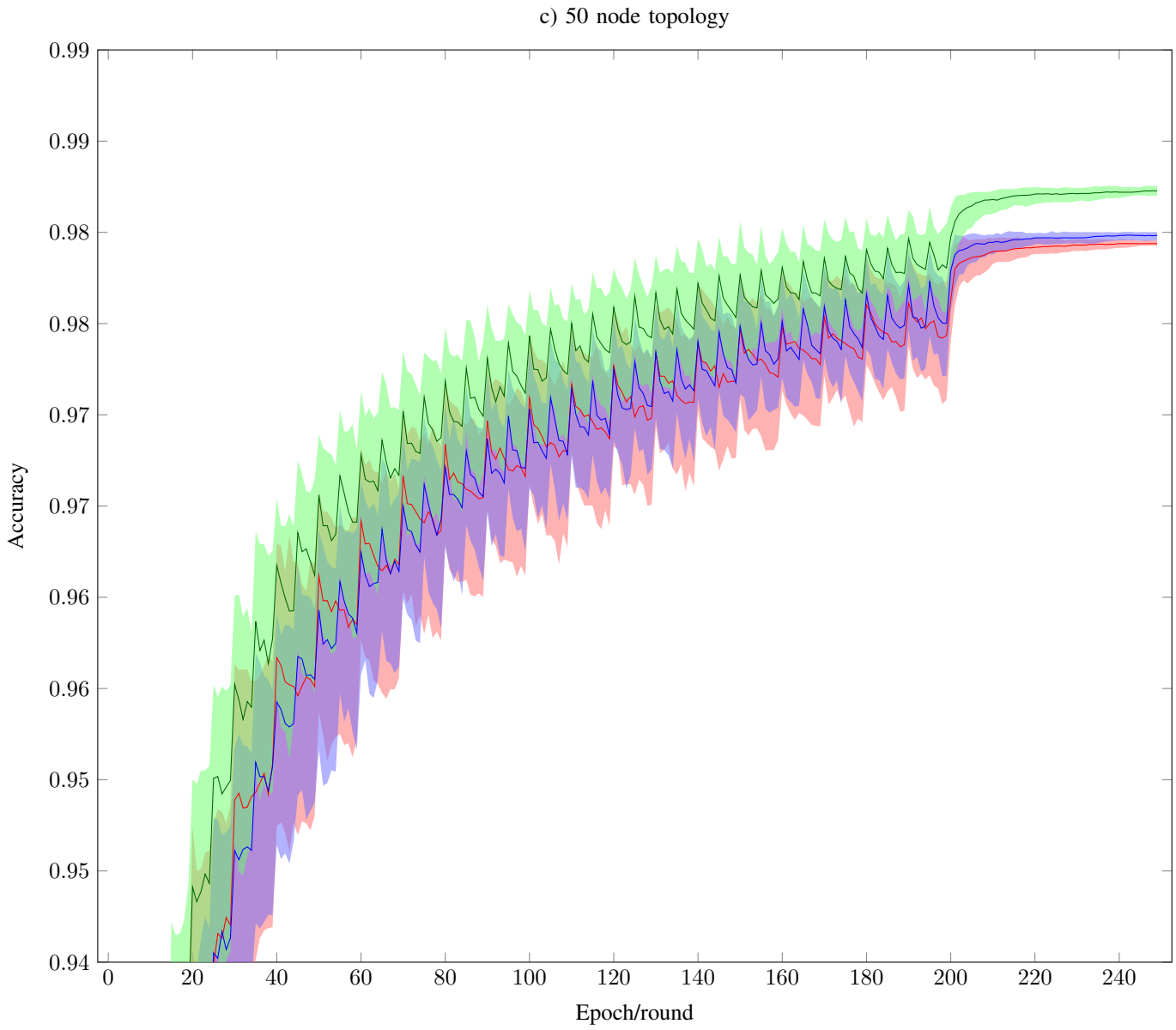



Fig. 6. Effect of increased update frequency $f \doteq \sqrt{n}$ and full message propagation, compared to baseline Delta Sum Learning (constant update frequency with default gossip cluster messaging behavior). While the update frequency by itself has little or no effect, the combination of both tweaks results in significantly better convergence.

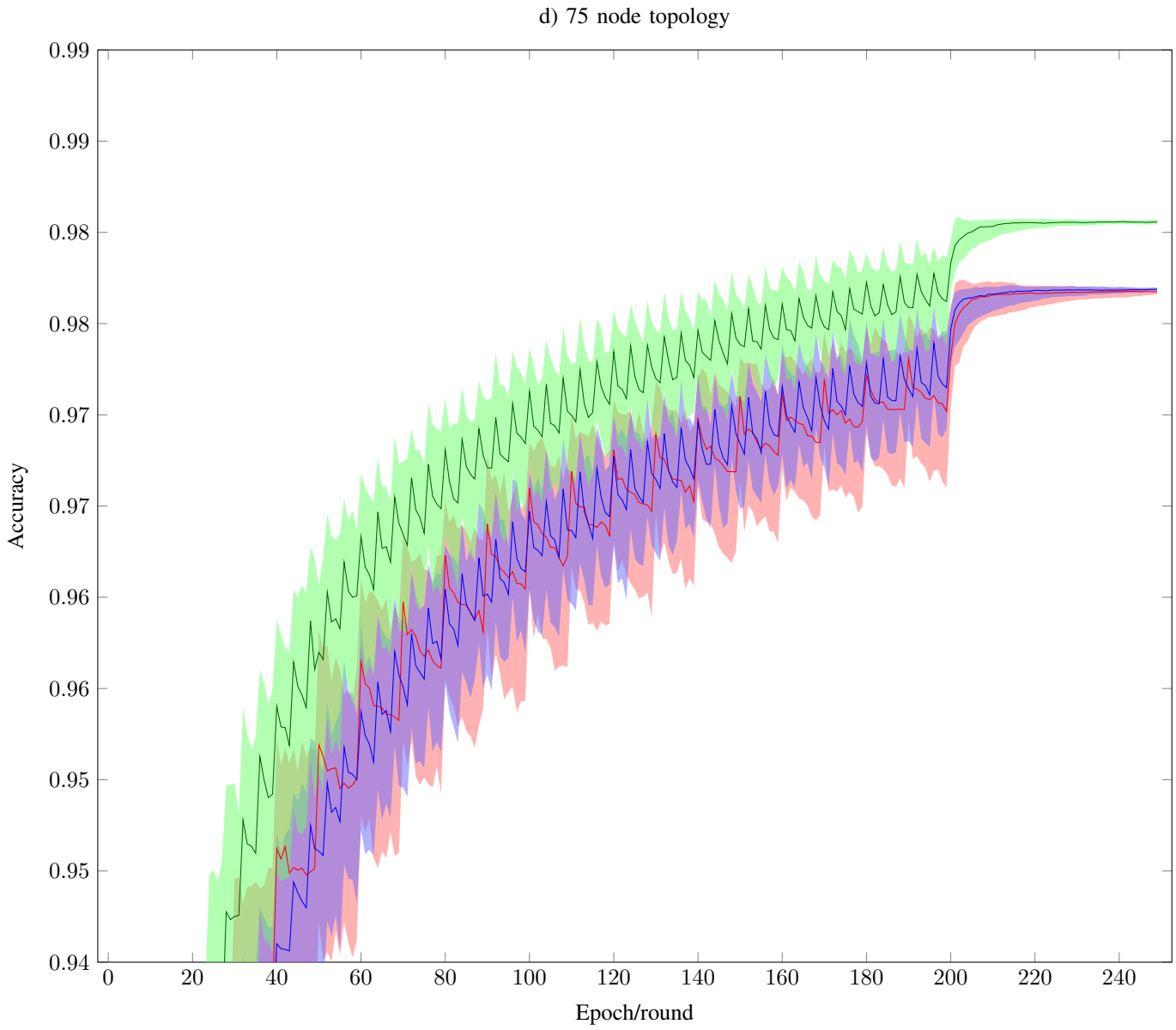



Fig. 7. Effect of increased update frequency $f \hat{=} \sqrt{n}$ and full message propagation, compared to baseline Delta Sum Learning (constant update frequency with default gossip cluster messaging behavior). While the update frequency by itself has little or no effect, the combination of both tweaks results in significantly better convergence.

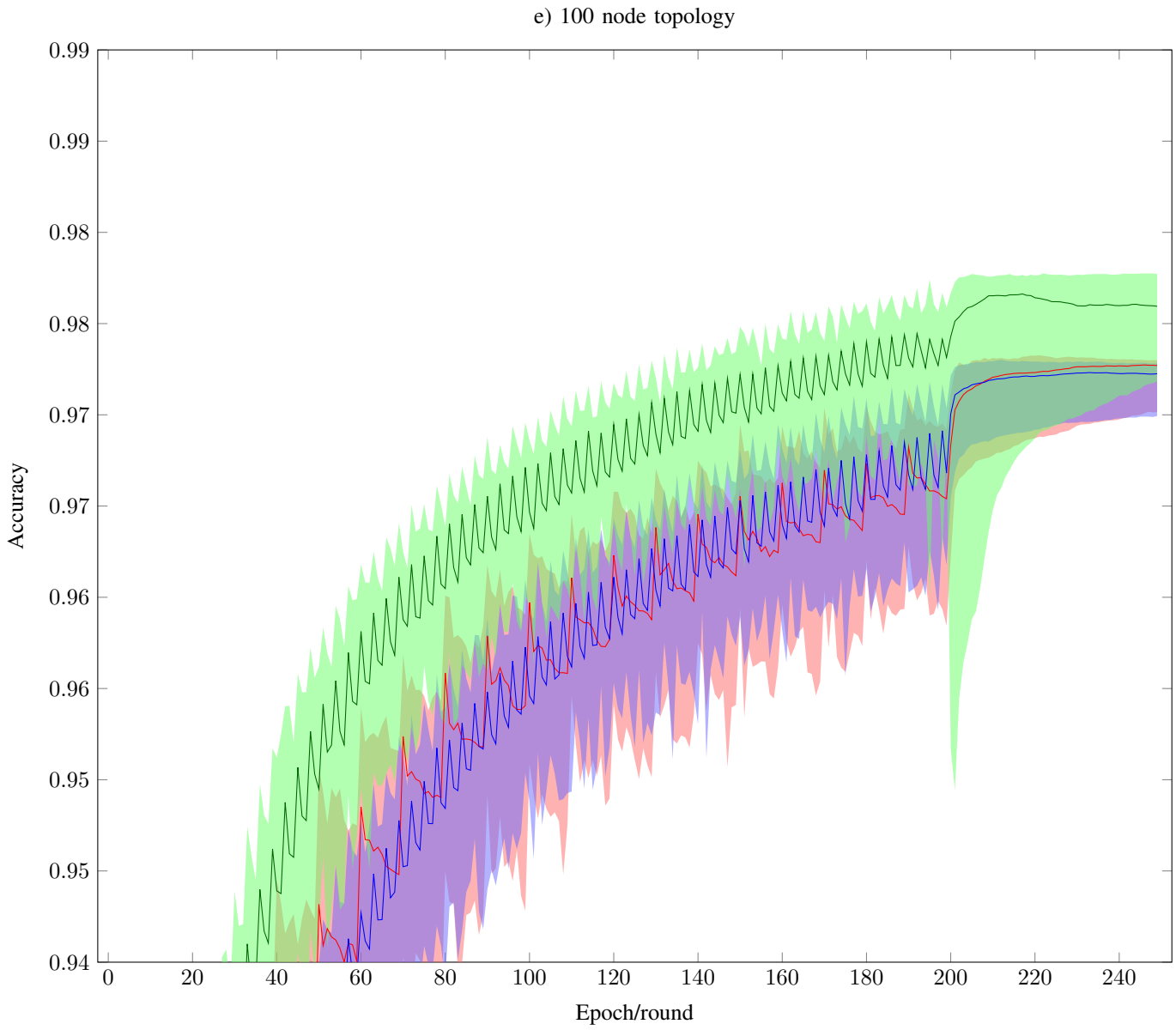



Fig. 8. Effect of increased update frequency $f \hat{=} \sqrt{n}$ and full message propagation, compared to baseline Delta Sum Learning (constant update frequency with default gossip cluster messaging behavior). While the update frequency by itself has little or no effect, the combination of both tweaks results in significantly better convergence.

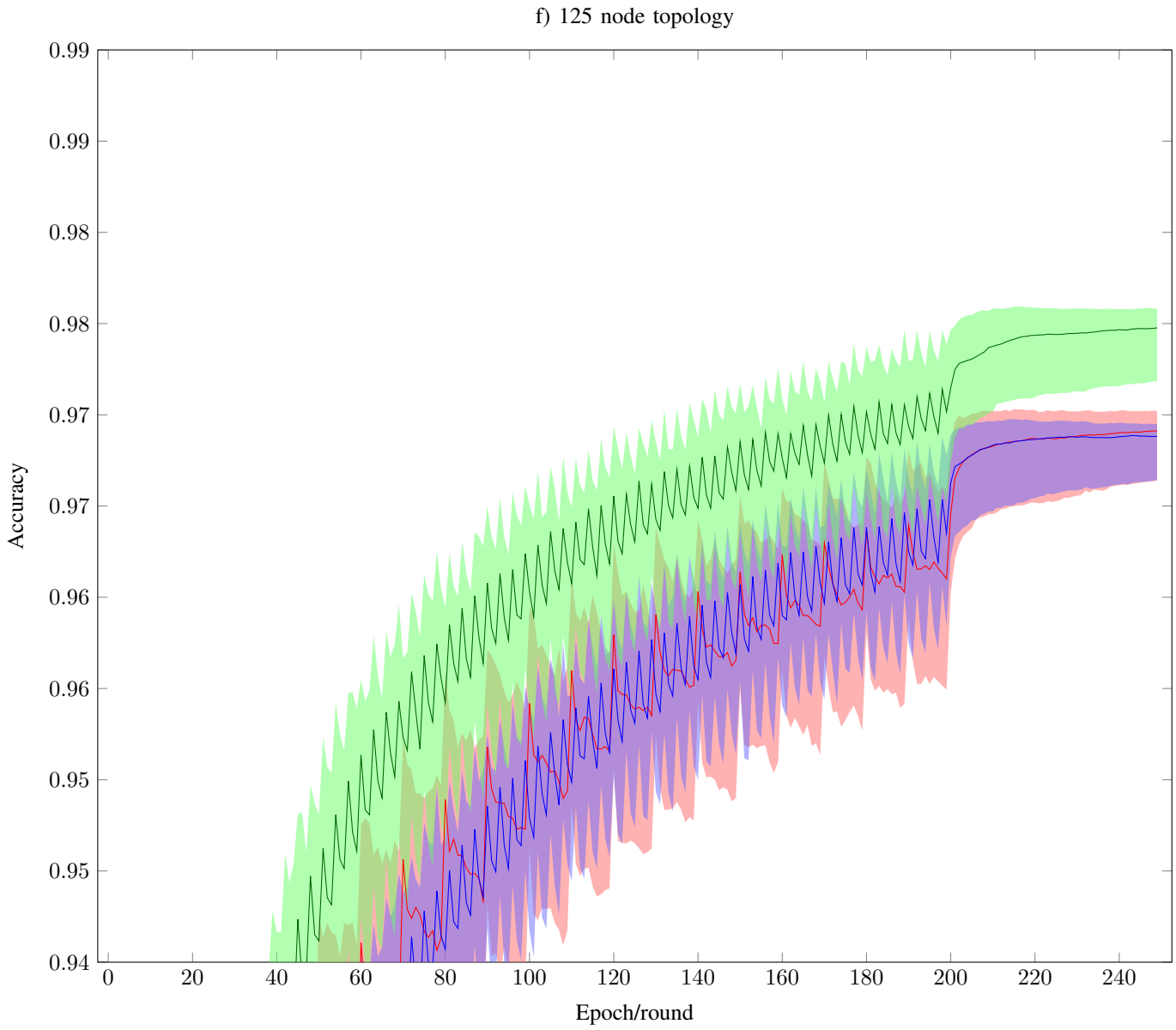



Fig. 9. Effect of increased update frequency $f \doteq \sqrt{n}$ and full message propagation, compared to baseline Delta Sum Learning (constant update frequency with default gossip cluster messaging behavior). While the update frequency by itself has little or no effect, the combination of both tweaks results in significantly better convergence.

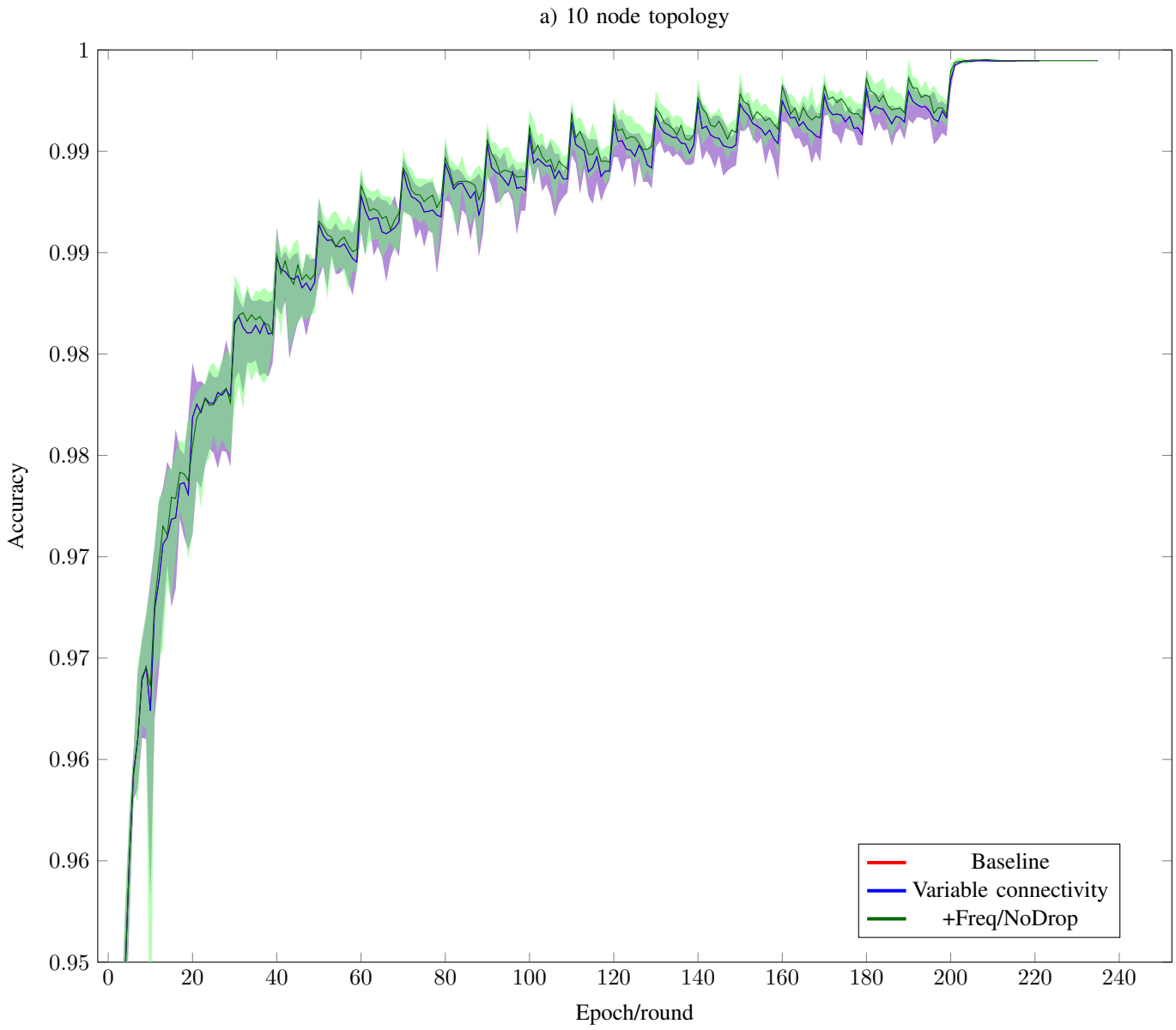



Fig. 10. Effect of variable connectivity and higher integration frequency/no gossip dropout compared to baseline Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role, potentially skewing or preventing global convergence. This effect is particularly visible in the 100 node topology.

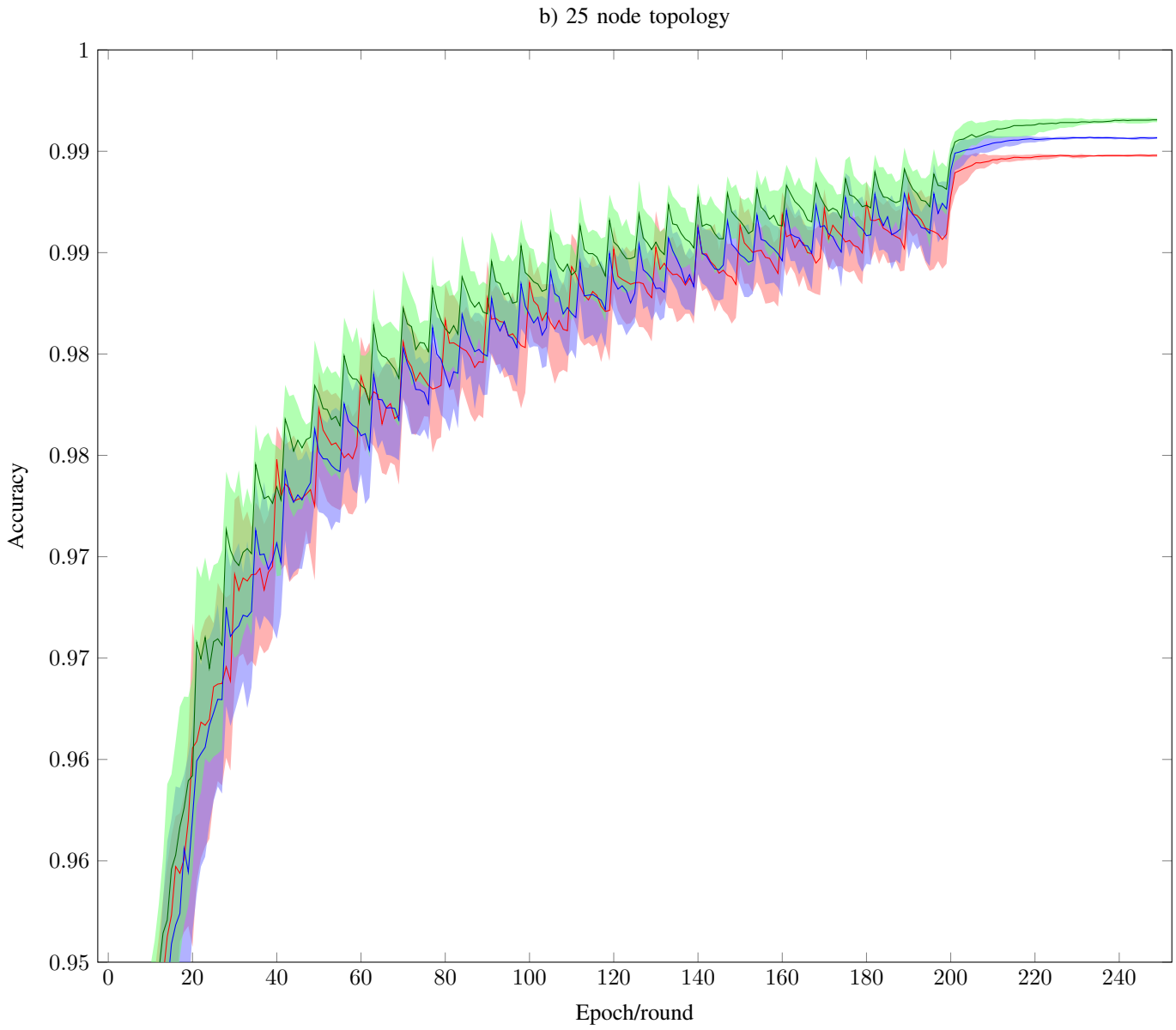



Fig. 11. Effect of variable connectivity and higher integration frequency/no gossip dropout compared to baseline Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role, potentially skewing or preventing global convergence. This effect is particularly visible in the 100 node topology.

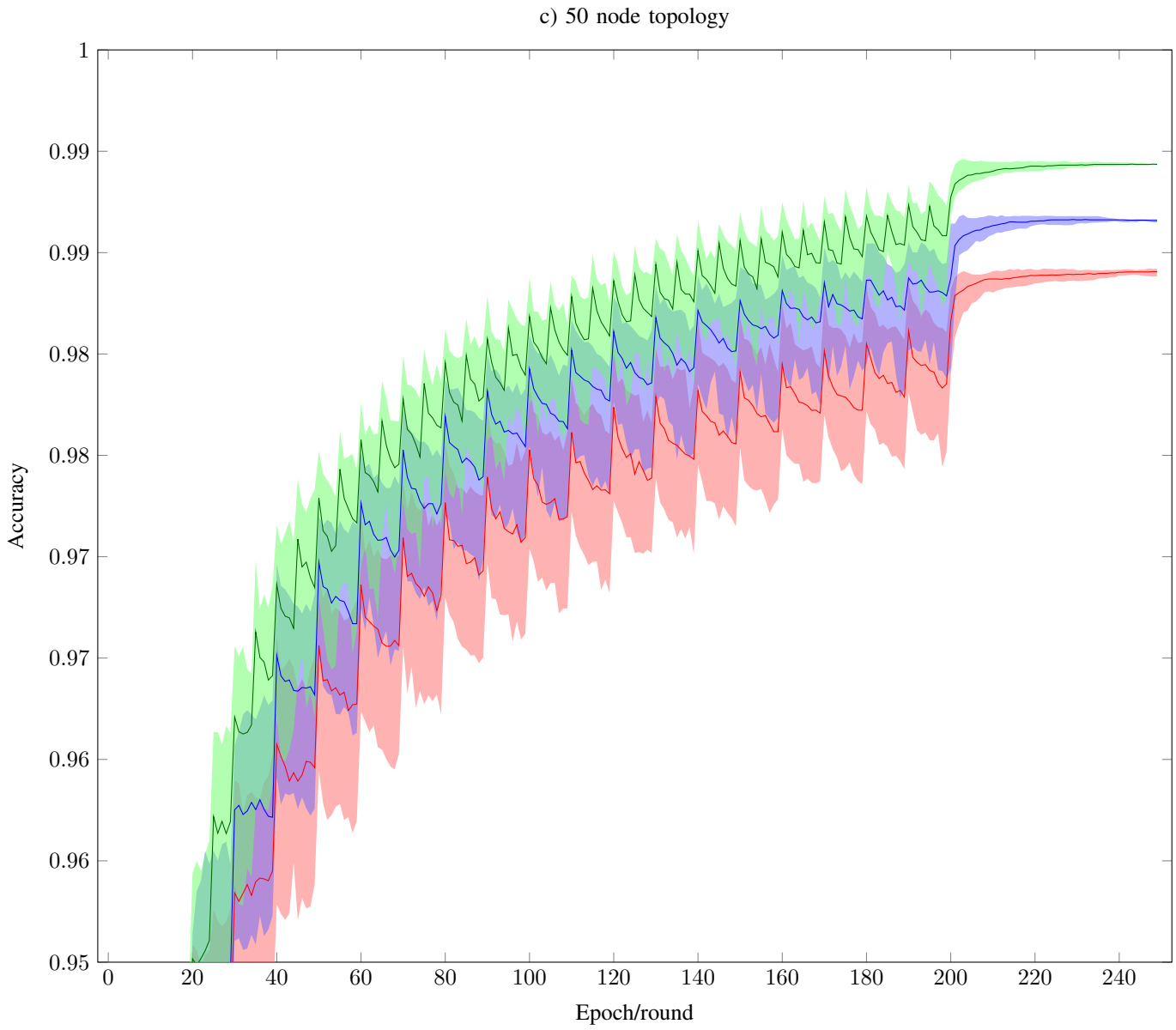



Fig. 12. Effect of variable connectivity and higher integration frequency/no gossip dropout compared to baseline Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role, potentially skewing or preventing global convergence. This effect is particularly visible in the 100 node topology.

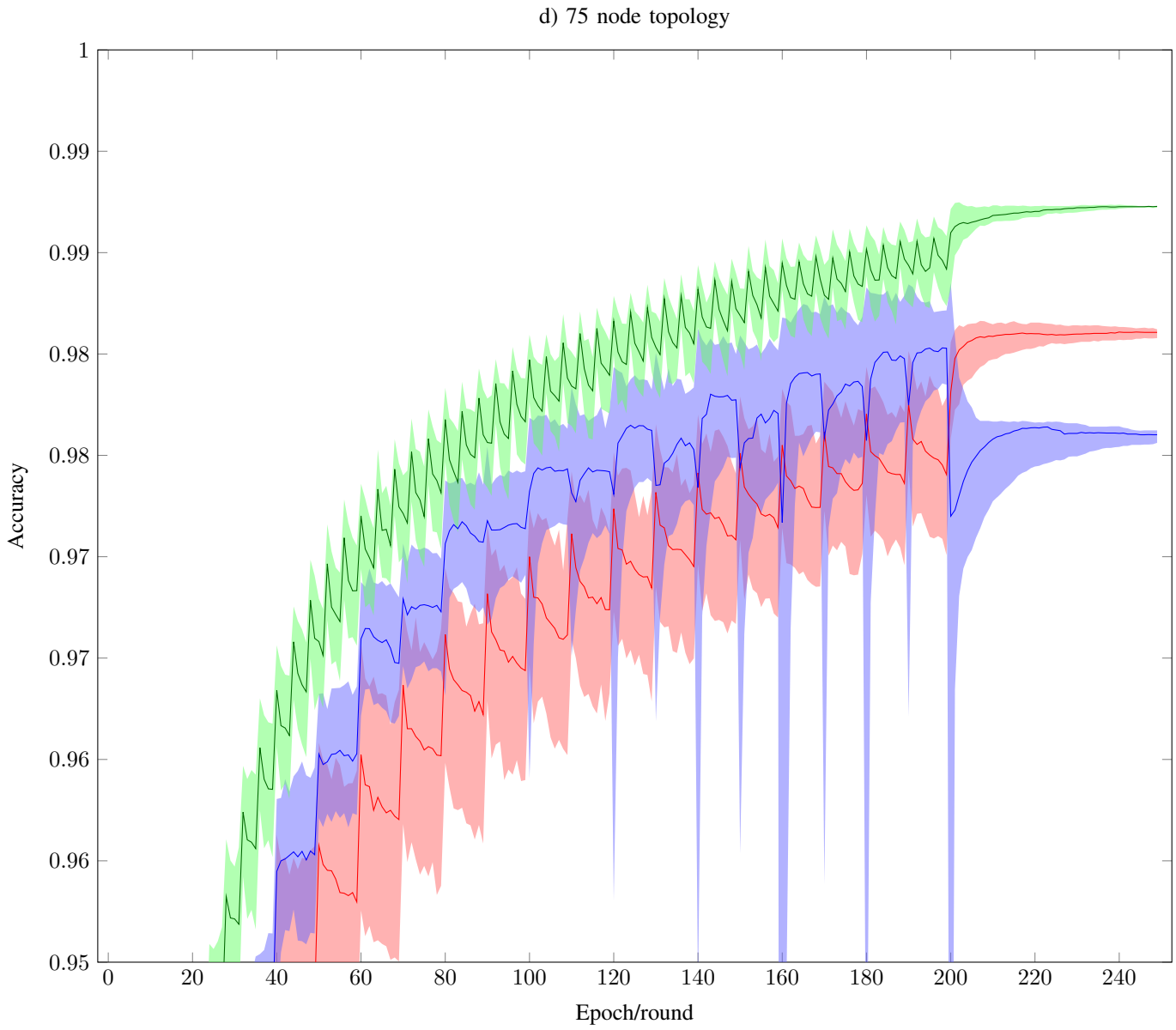



Fig. 13. Effect of variable connectivity and higher integration frequency/no gossip dropout compared to baseline Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role, potentially skewing or preventing global convergence. This effect is particularly visible in the 100 node topology.

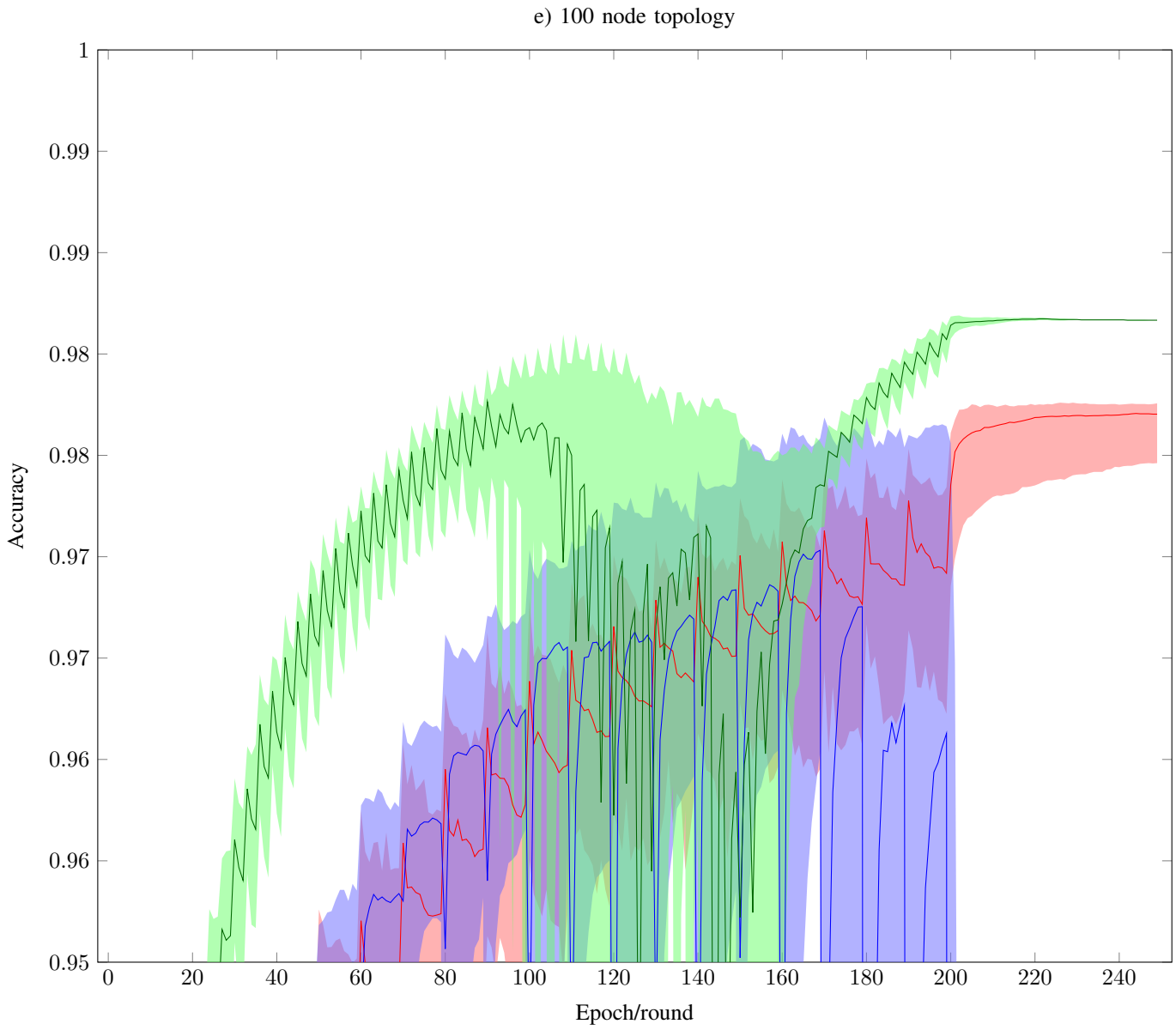



Fig. 14. Effect of variable connectivity and higher integration frequency/no gossip dropout compared to baseline Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role, potentially skewing or preventing global convergence. This effect is particularly visible in the 100 node topology.

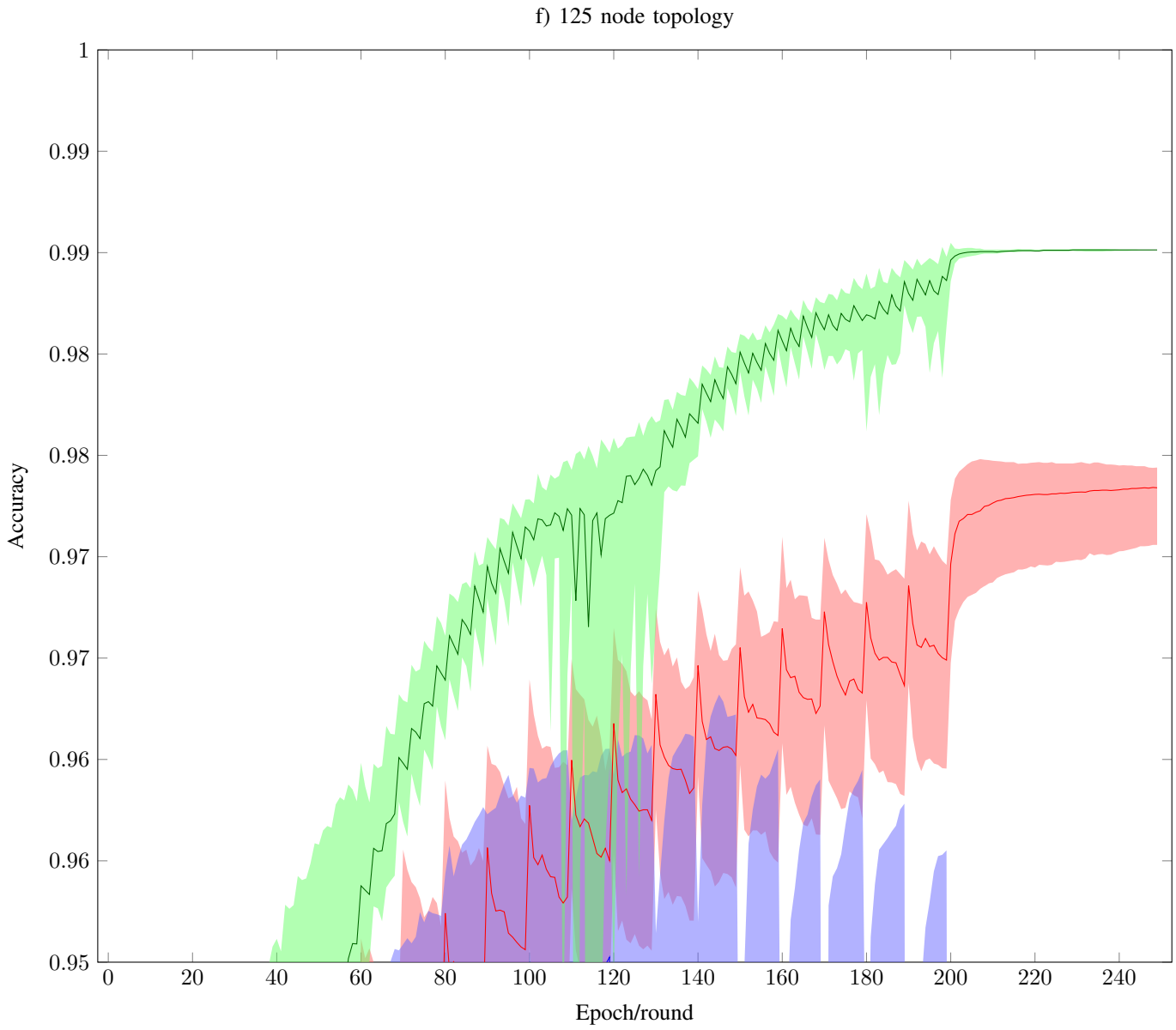



Fig. 15. Effect of variable connectivity and higher integration frequency/no gossip dropout compared to baseline Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role, potentially skewing or preventing global convergence. This effect is particularly visible in the 100 node topology.

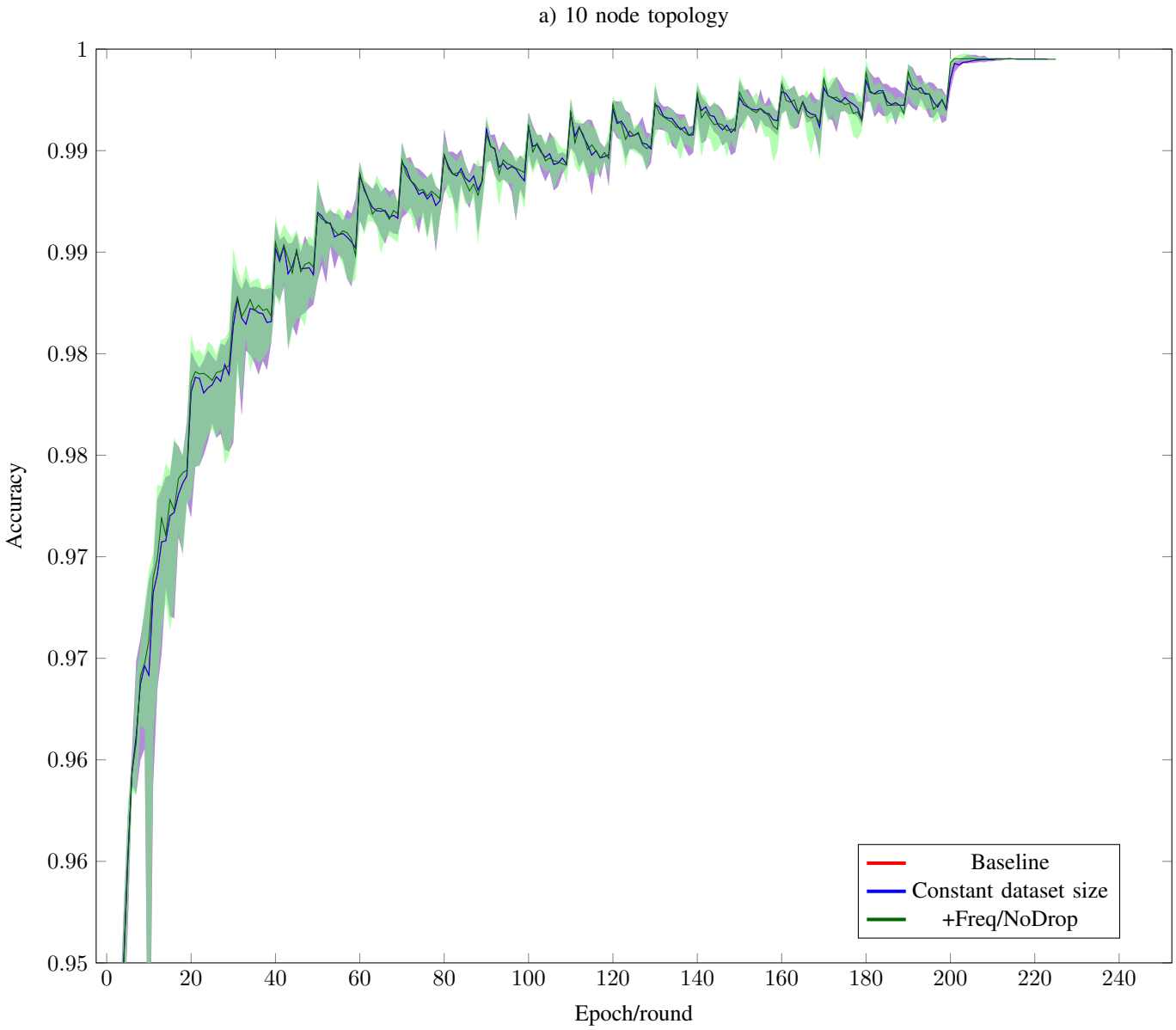



Fig. 16. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout.

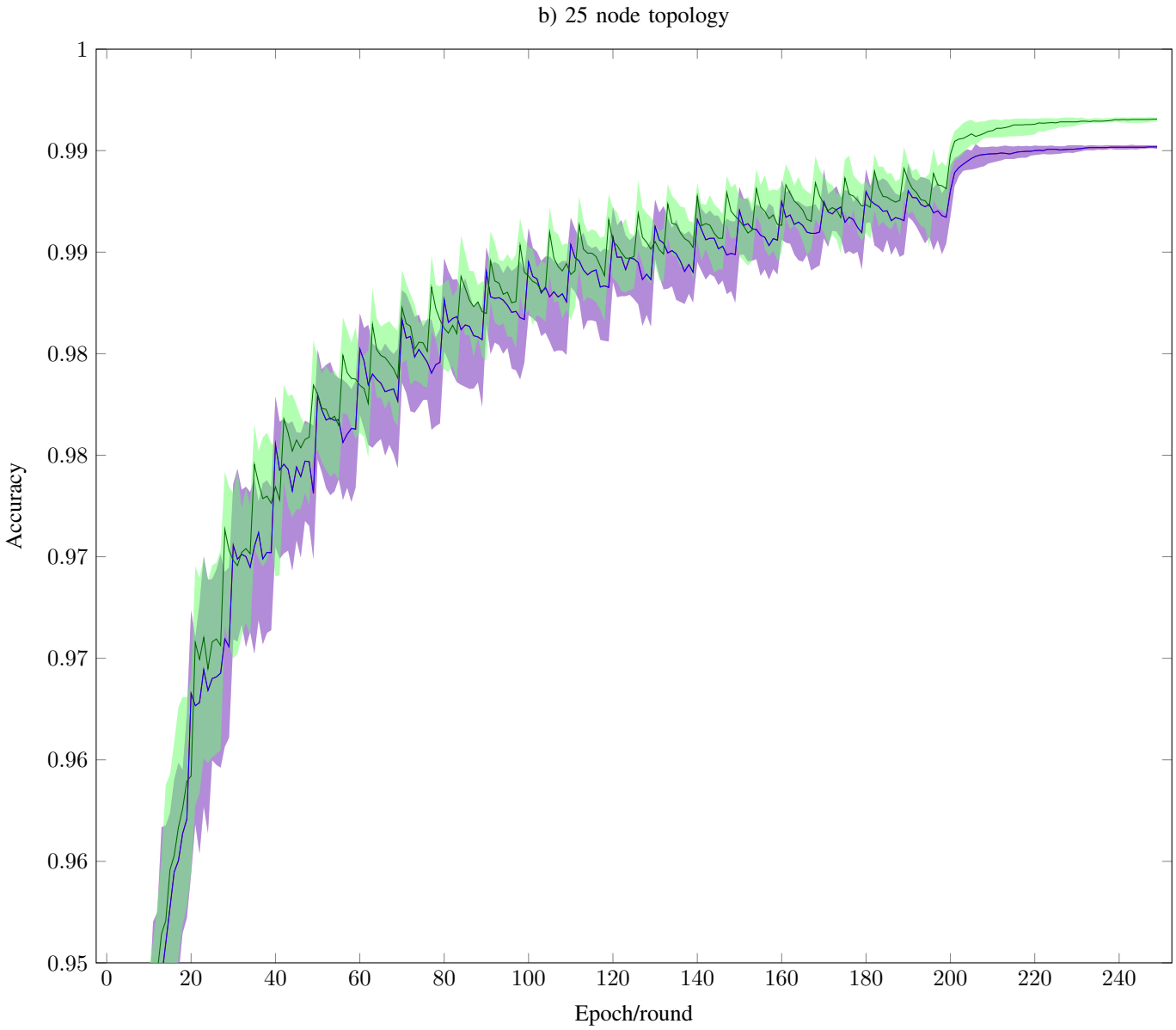



Fig. 17. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout.

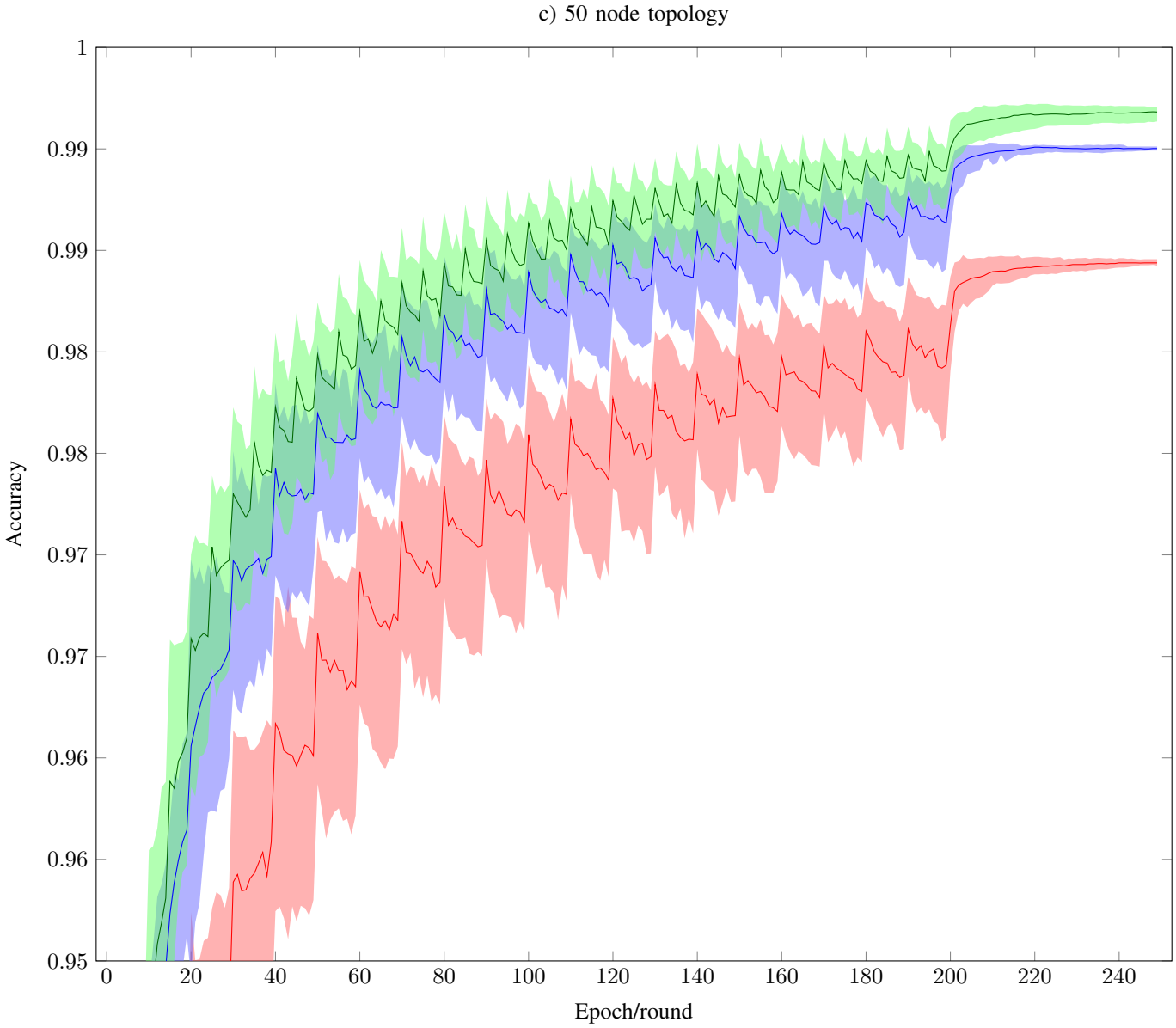



Fig. 18. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout.

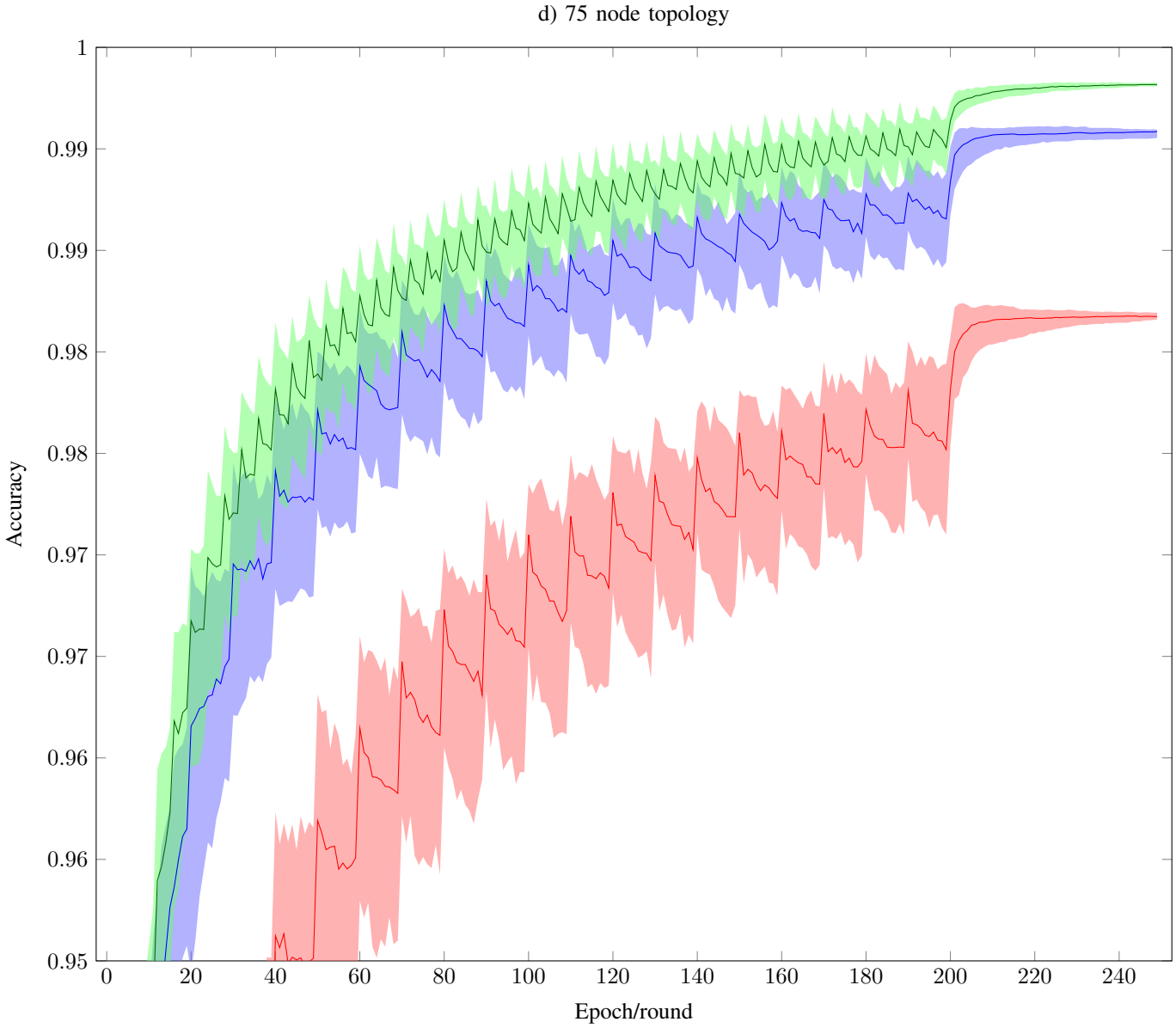



Fig. 19. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout.

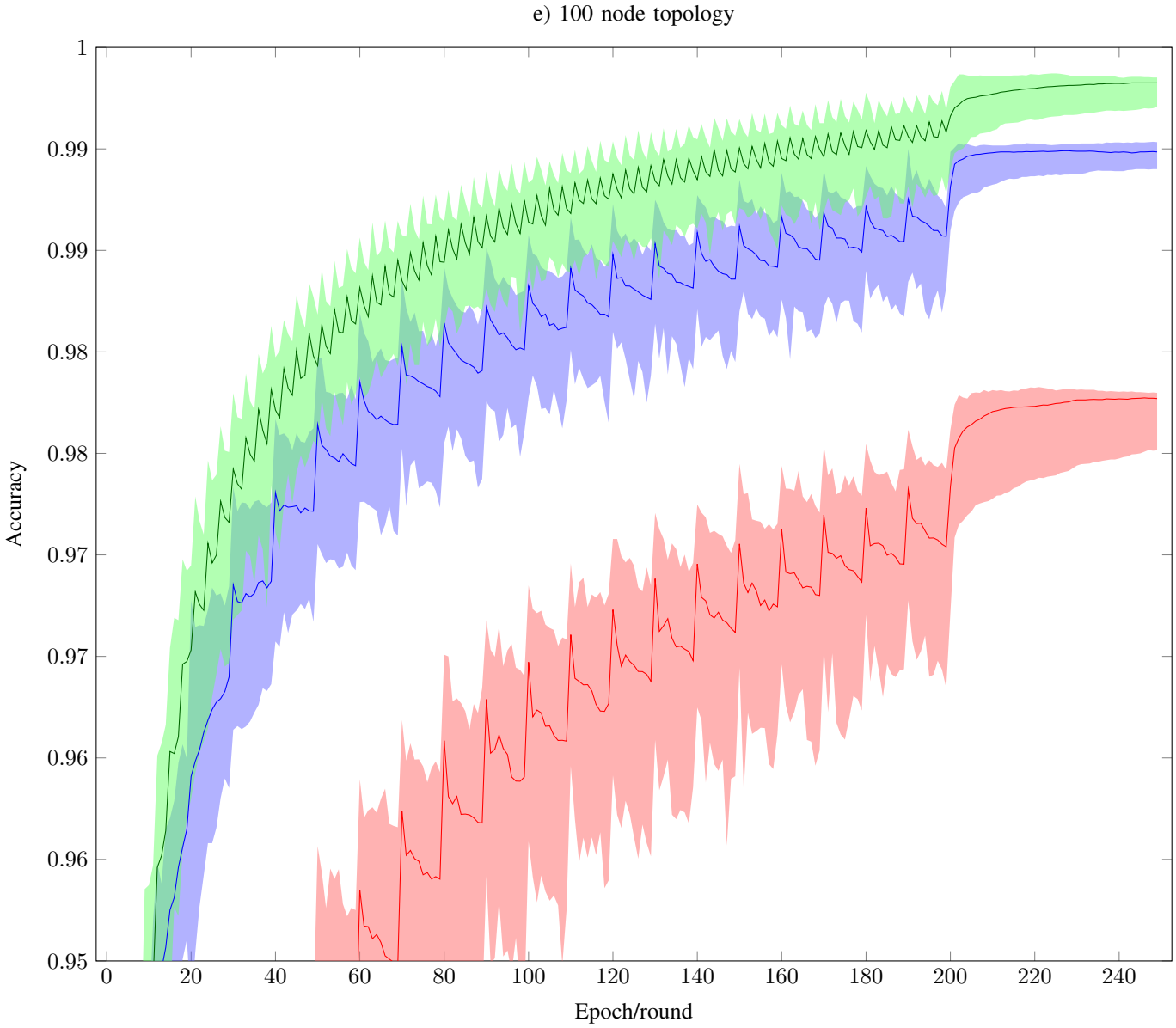



Fig. 20. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout.

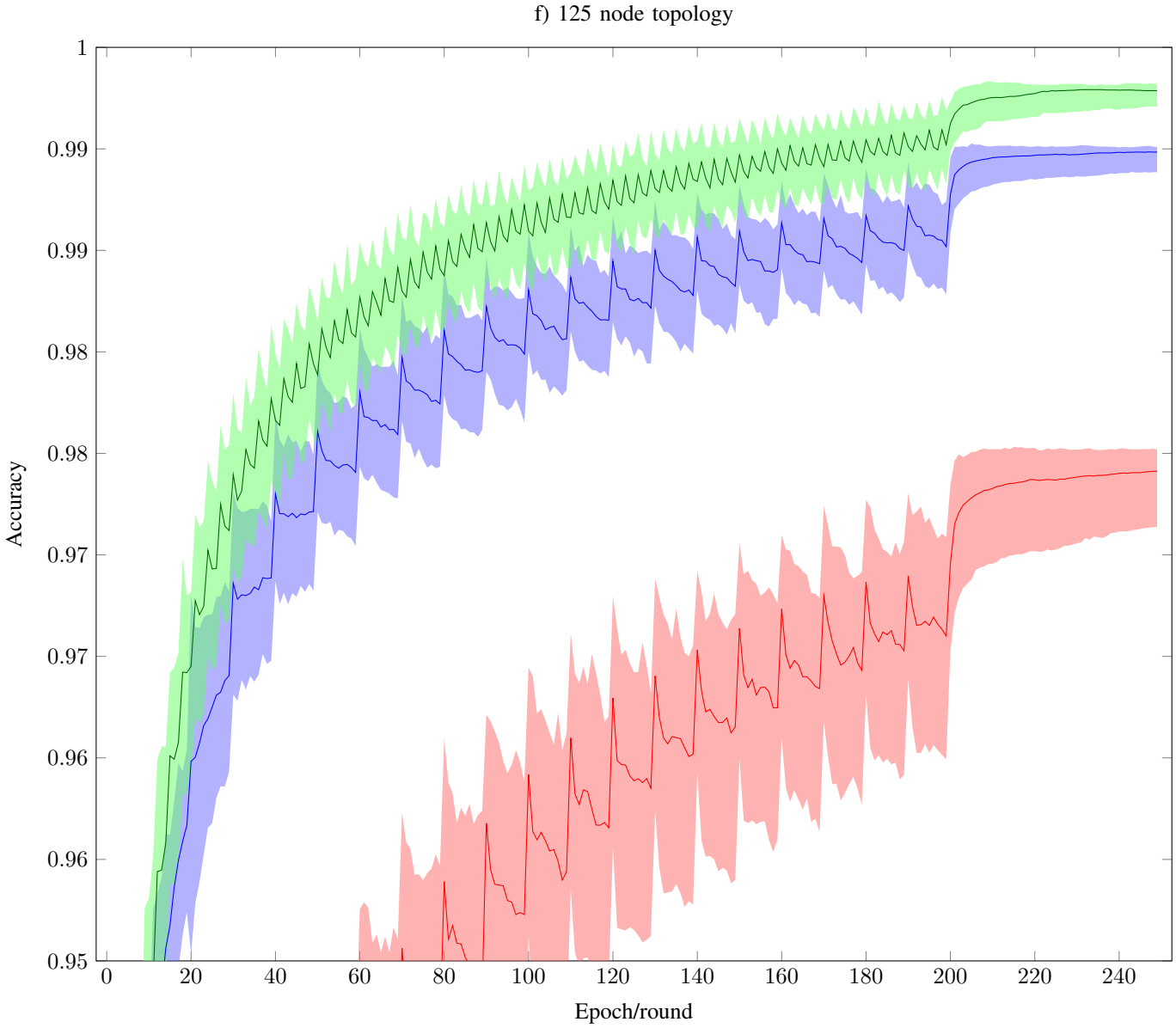



Fig. 21. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout.

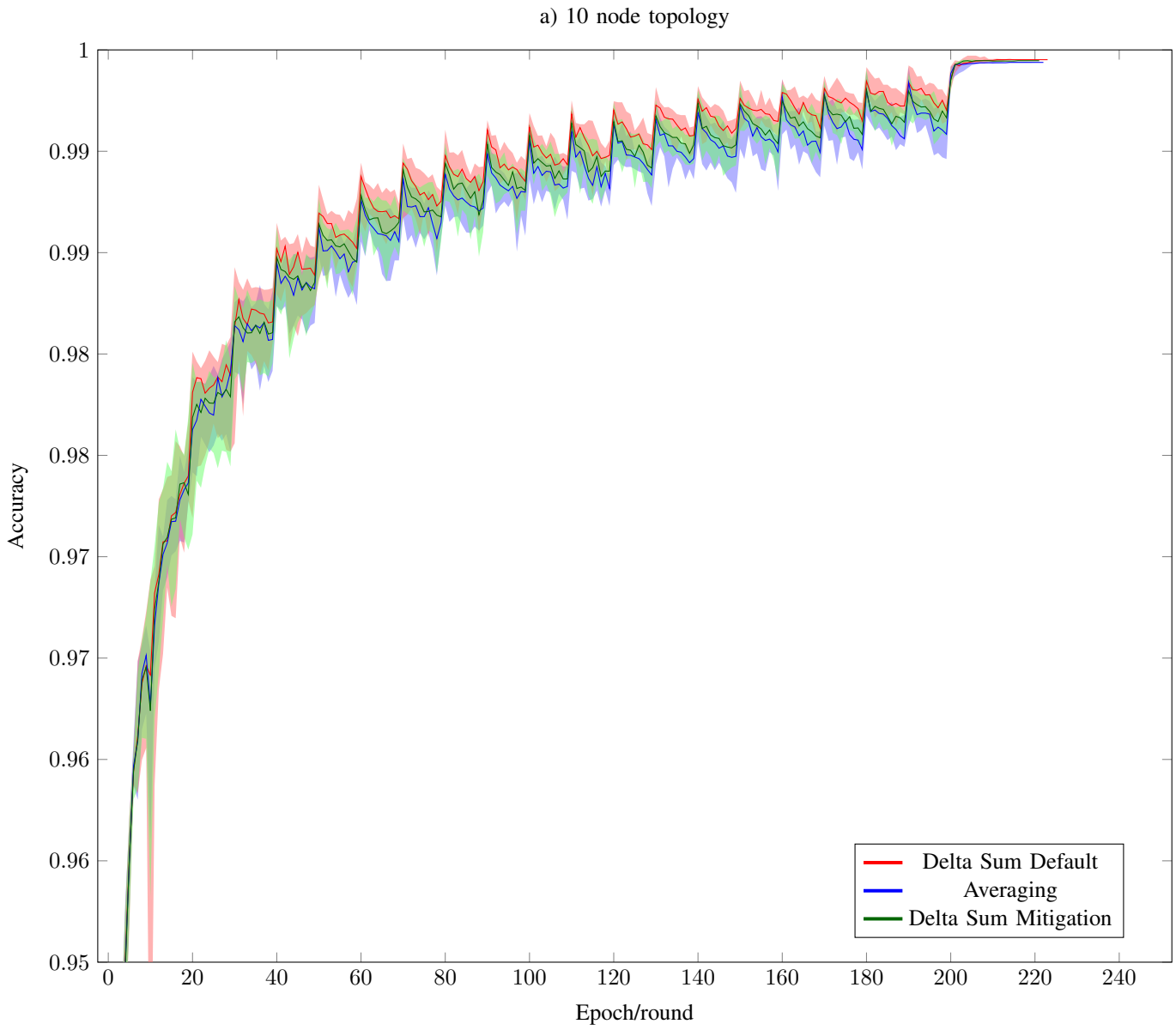



Fig. 22. Training performance of Delta Sum Learning with and without mitigation measures compared to default Averaging. While the mitigation measures for antagonistic topologies and training scenarios have a slight impact on Delta Sum Learning in this "well behaved" configuration, both versions perform better than Averaging, with an increasing accuracy gap as topologies grow.

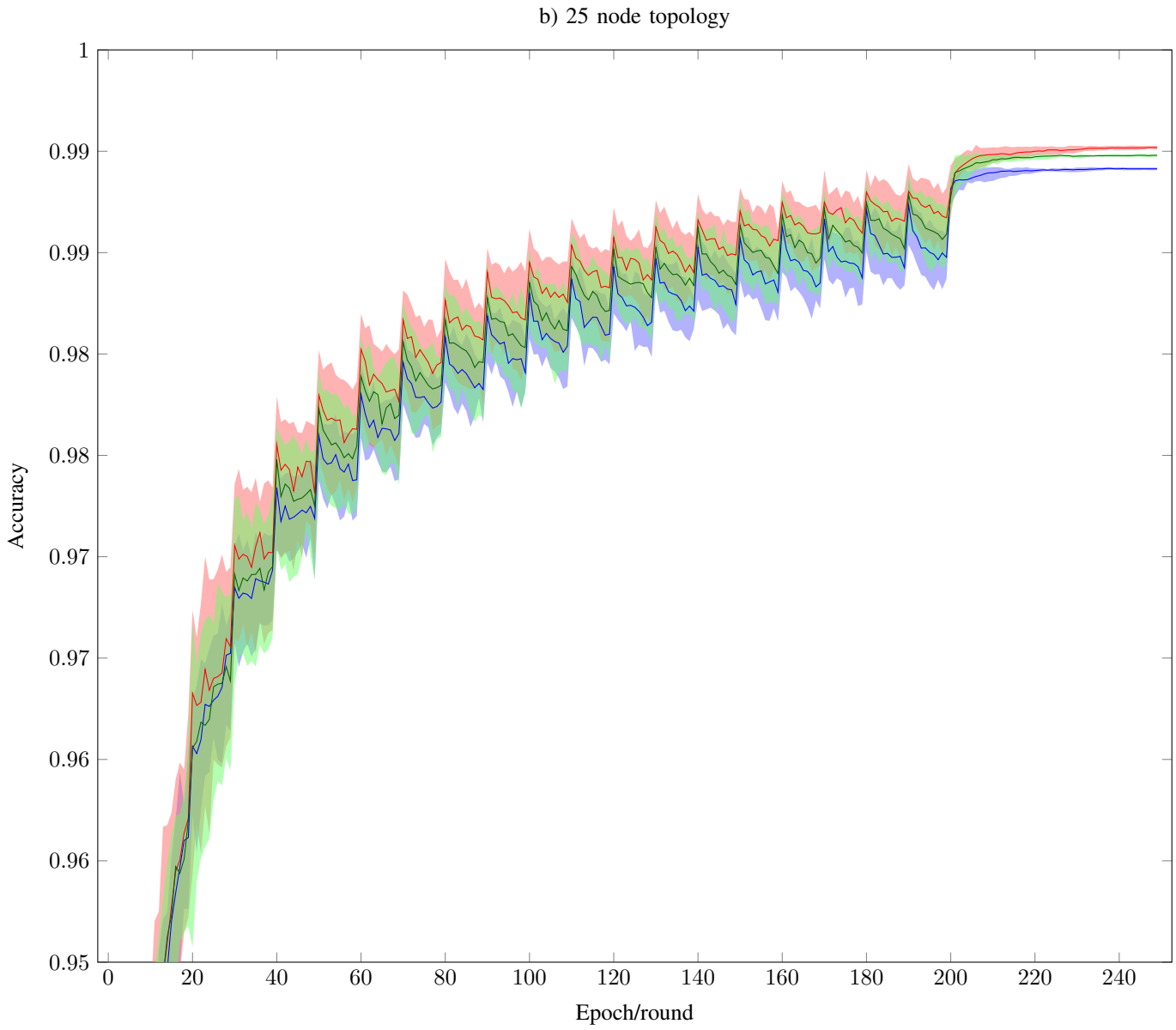



Fig. 23. Training performance of Delta Sum Learning with and without mitigation measures compared to default Averaging. While the mitigation measures for antagonistic topologies and training scenarios have a slight impact on Delta Sum Learning in this "well behaved" configuration, both versions perform better than Averaging, with an increasing accuracy gap as topologies grow.

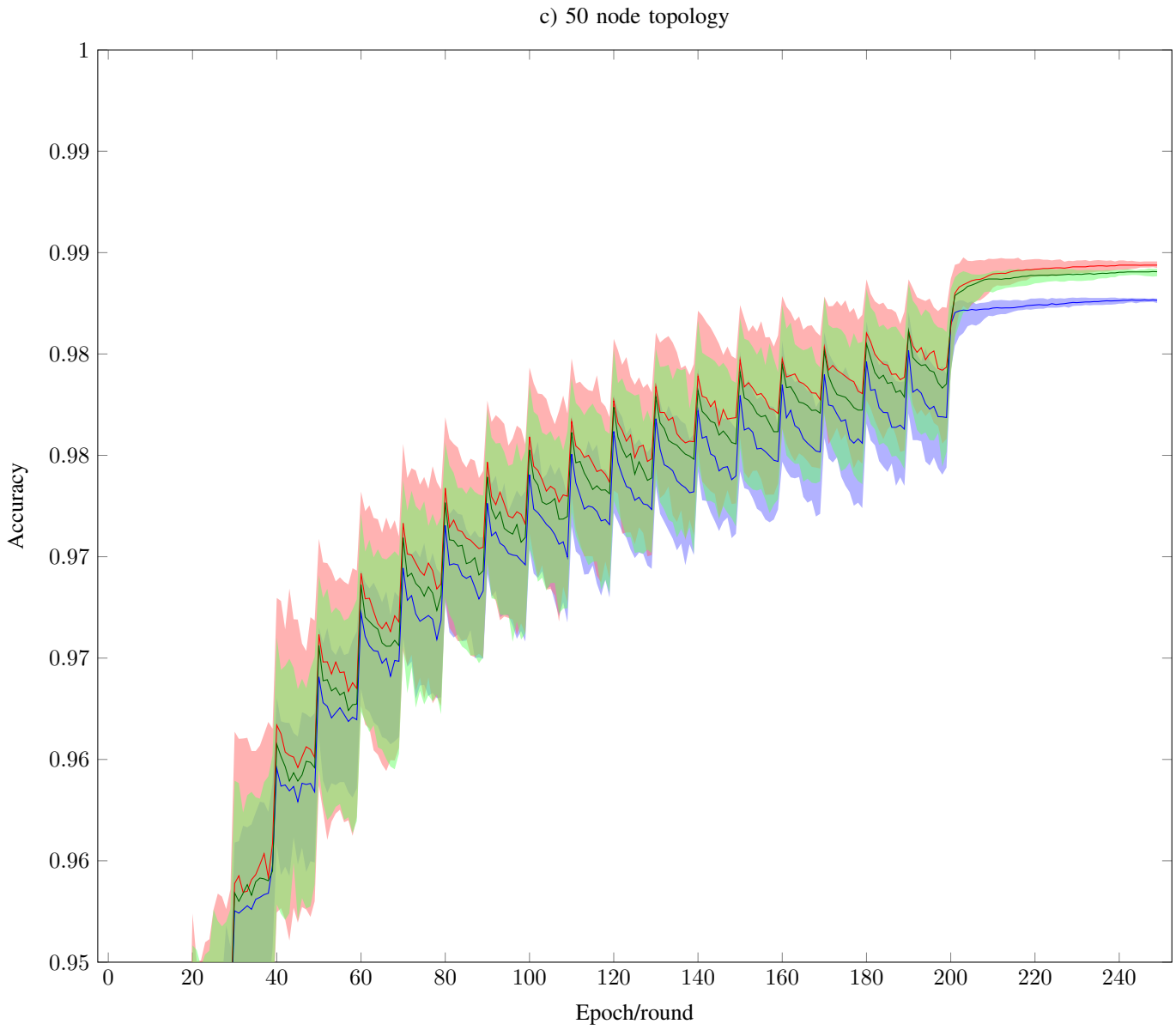



Fig. 24. Training performance of Delta Sum Learning with and without mitigation measures compared to default Averaging. While the mitigation measures for antagonistic topologies and training scenarios have a slight impact on Delta Sum Learning in this "well behaved" configuration, both versions perform better than Averaging, with an increasing accuracy gap as topologies grow.

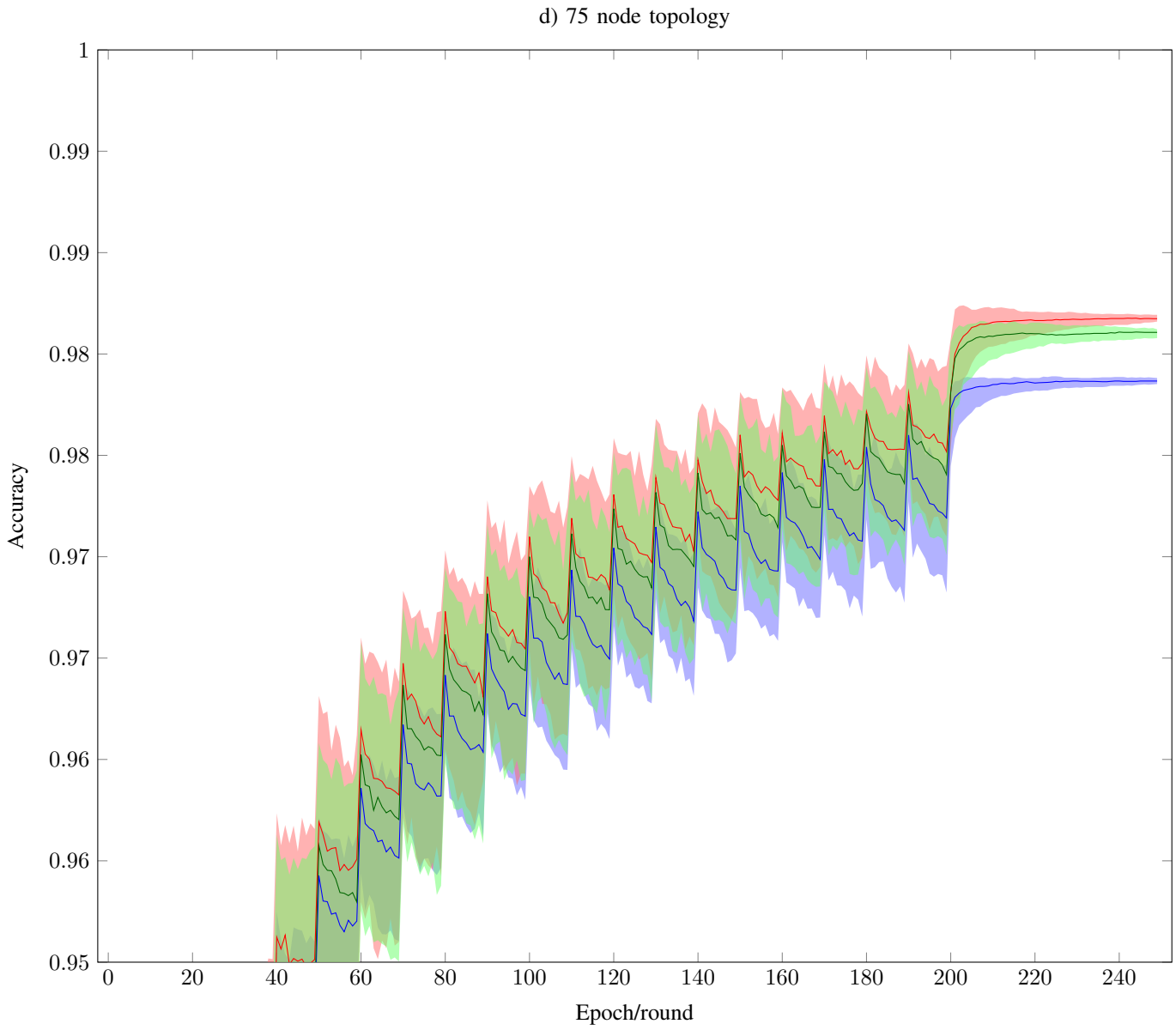



Fig. 25. Training performance of Delta Sum Learning with and without mitigation measures compared to default Averaging. While the mitigation measures for antagonistic topologies and training scenarios have a slight impact on Delta Sum Learning in this "well behaved" configuration, both versions perform better than Averaging, with an increasing accuracy gap as topologies grow.

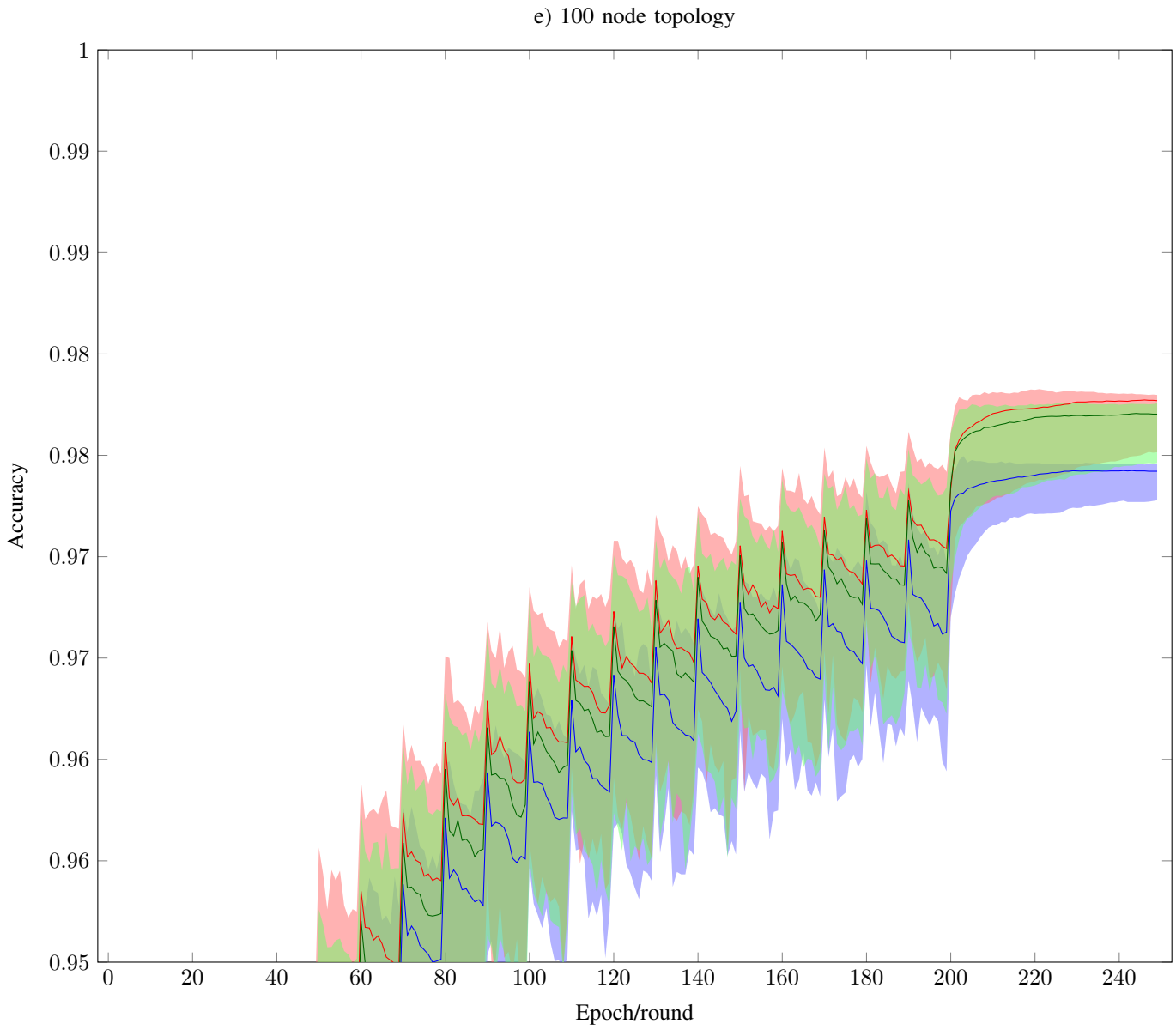



Fig. 26. Training performance of Delta Sum Learning with and without mitigation measures compared to default Averaging. While the mitigation measures for antagonistic topologies and training scenarios have a slight impact on Delta Sum Learning in this "well behaved" configuration, both versions perform better than Averaging, with an increasing accuracy gap as topologies grow.

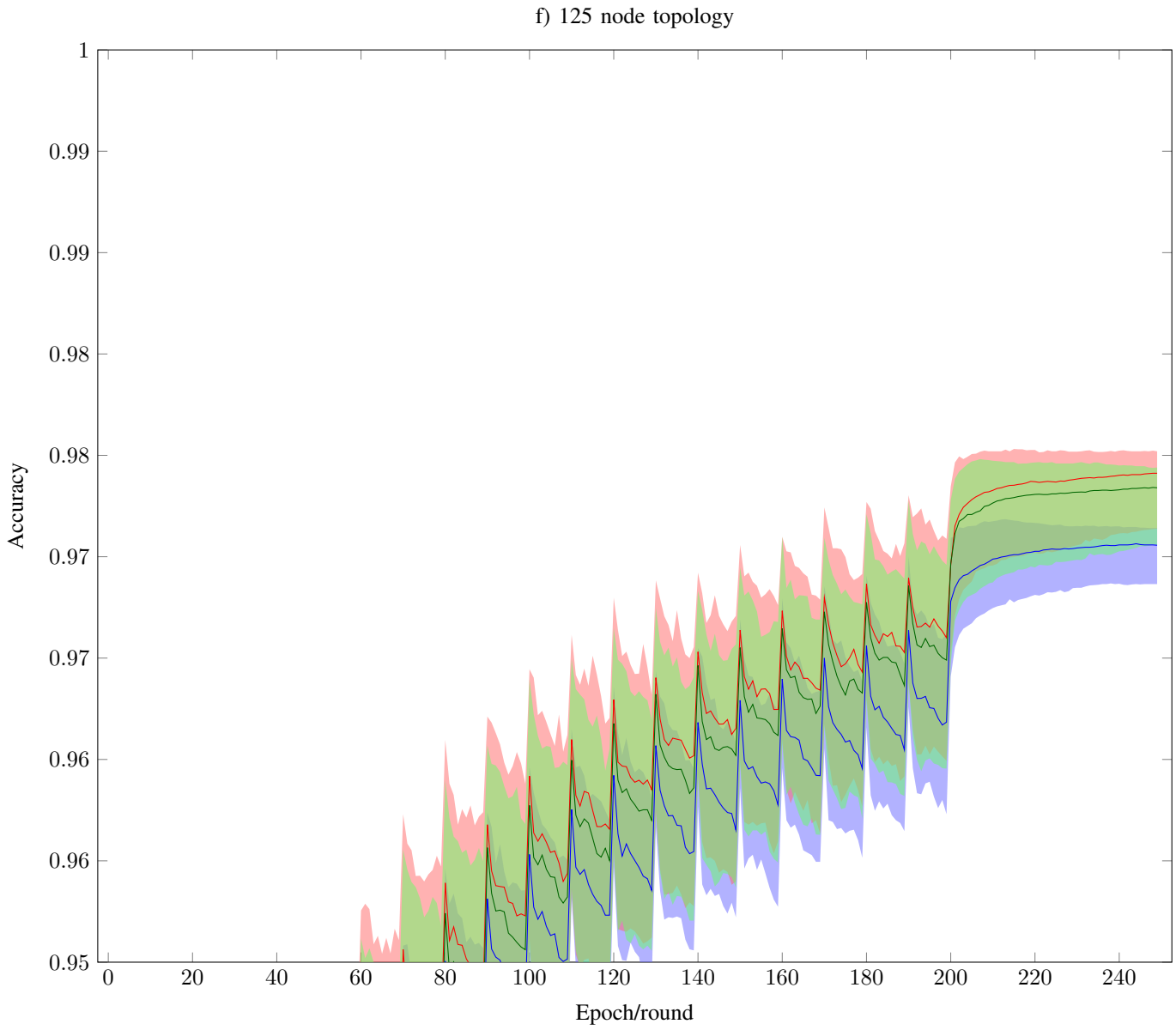



Fig. 27. Training performance of Delta Sum Learning with and without mitigation measures compared to default Averaging. While the mitigation measures for antagonistic topologies and training scenarios have a slight impact on Delta Sum Learning in this "well behaved" configuration, both versions perform better than Averaging, with an increasing accuracy gap as topologies grow.

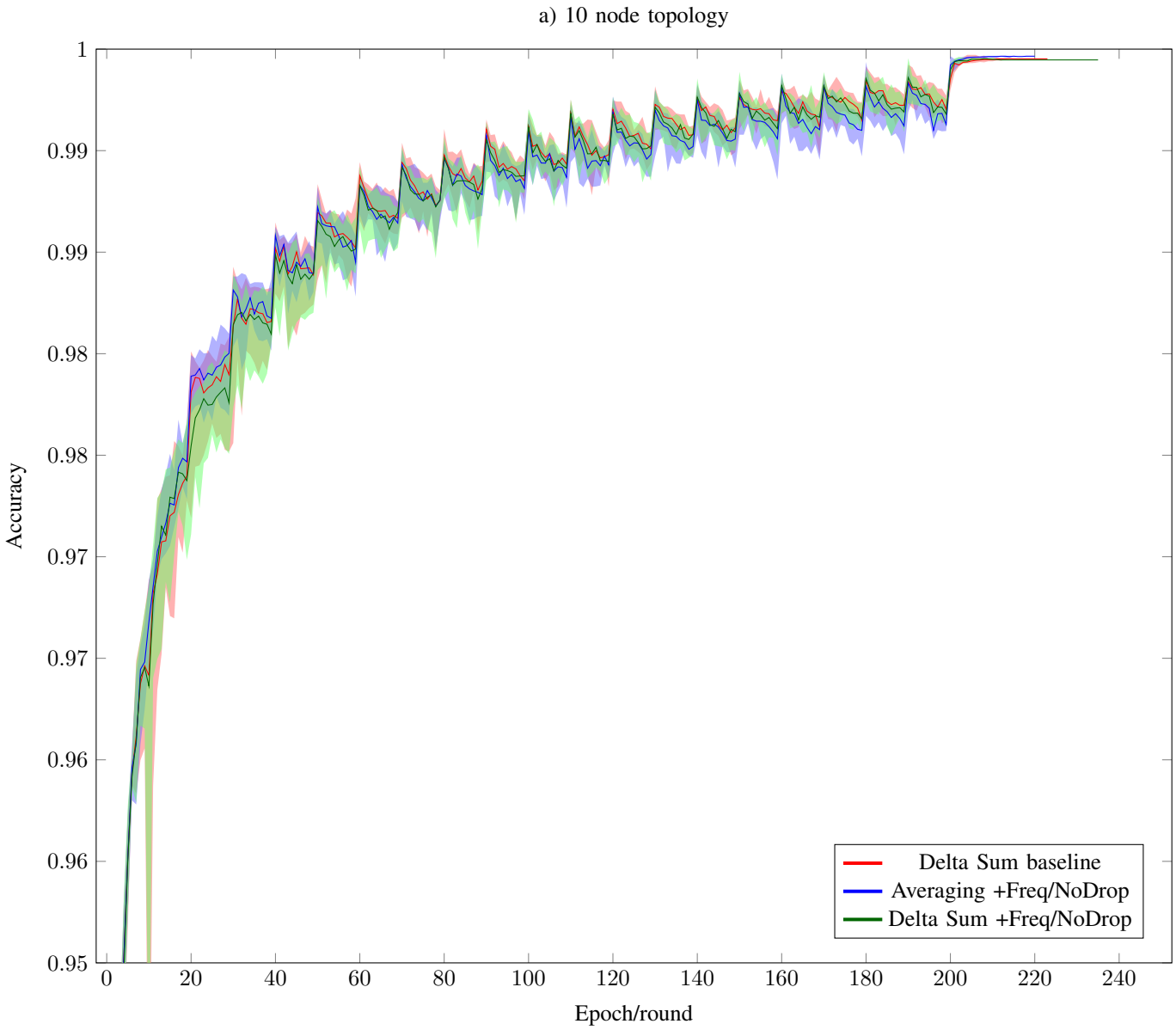



Fig. 28. Effect of increased update frequency $f \sim \sqrt{n}$ and full message propagation on both Delta Sum Learning and Averaging, compared to baseline Delta Sum Learning.

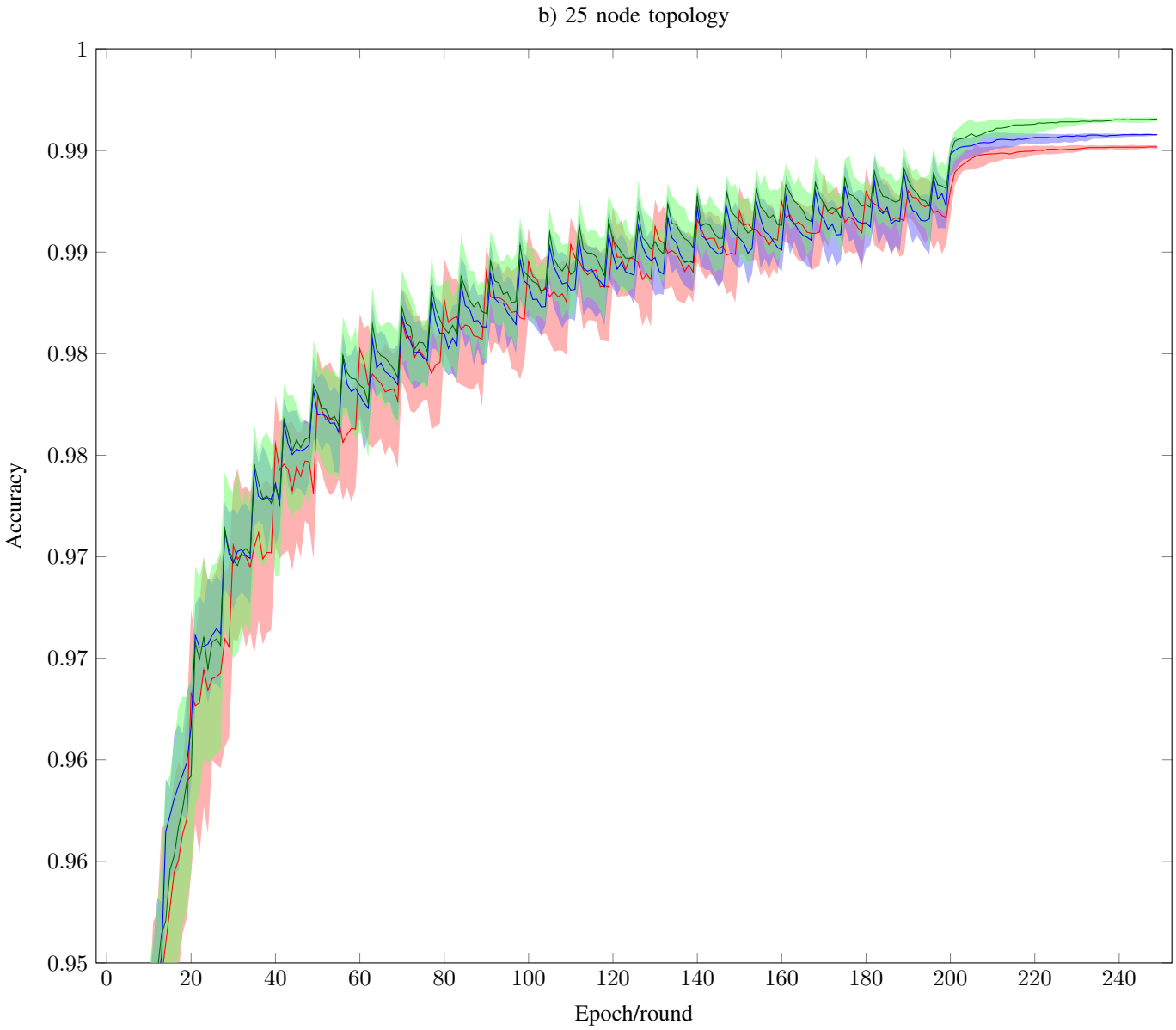



Fig. 29. Effect of increased update frequency $f \sim \sqrt{n}$ and full message propagation on both Delta Sum Learning and Averaging, compared to baseline Delta Sum Learning.

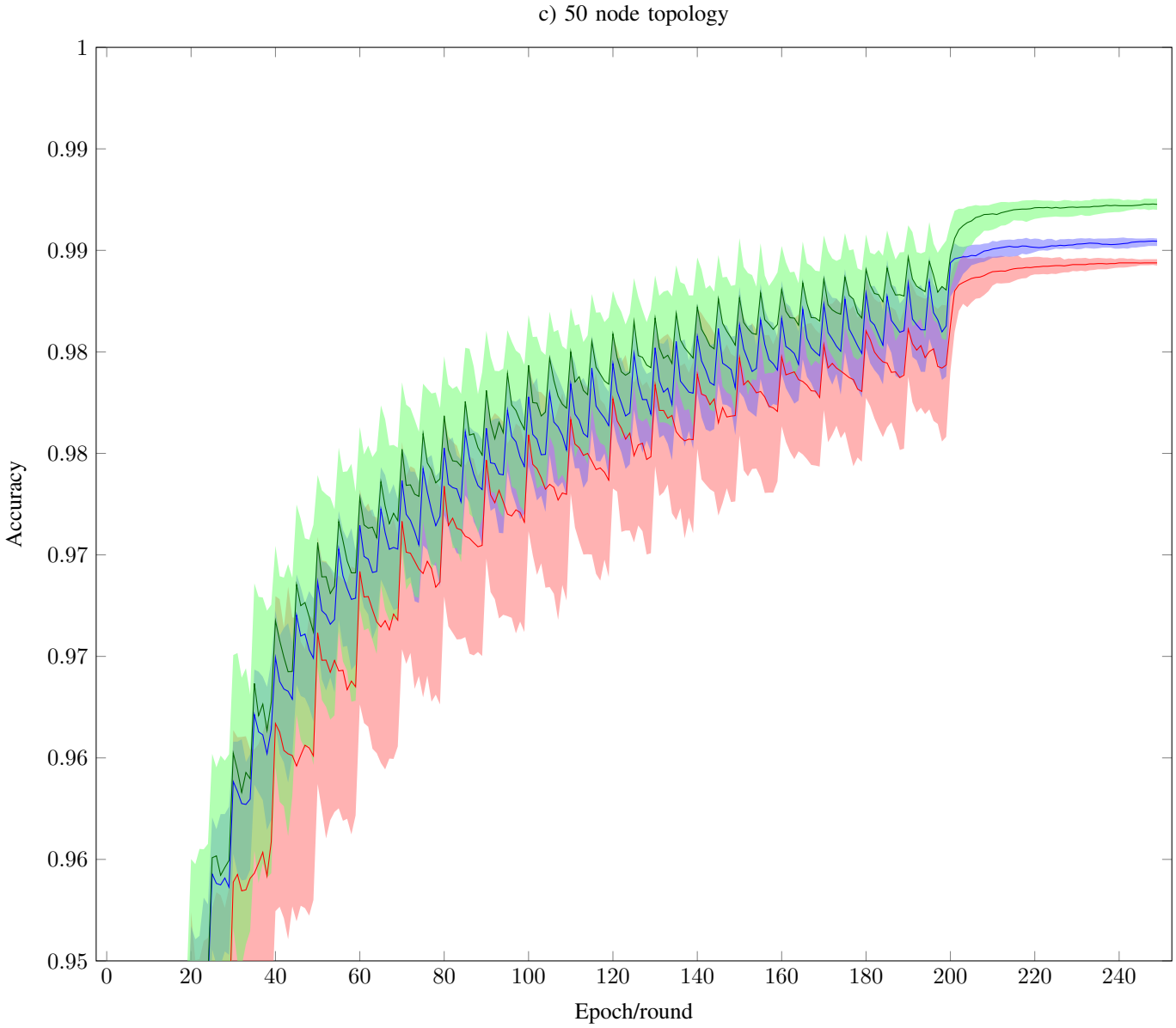



Fig. 30. Effect of increased update frequency $f \sim \sqrt{n}$ and full message propagation on both Delta Sum Learning and Averaging, compared to baseline Delta Sum Learning.

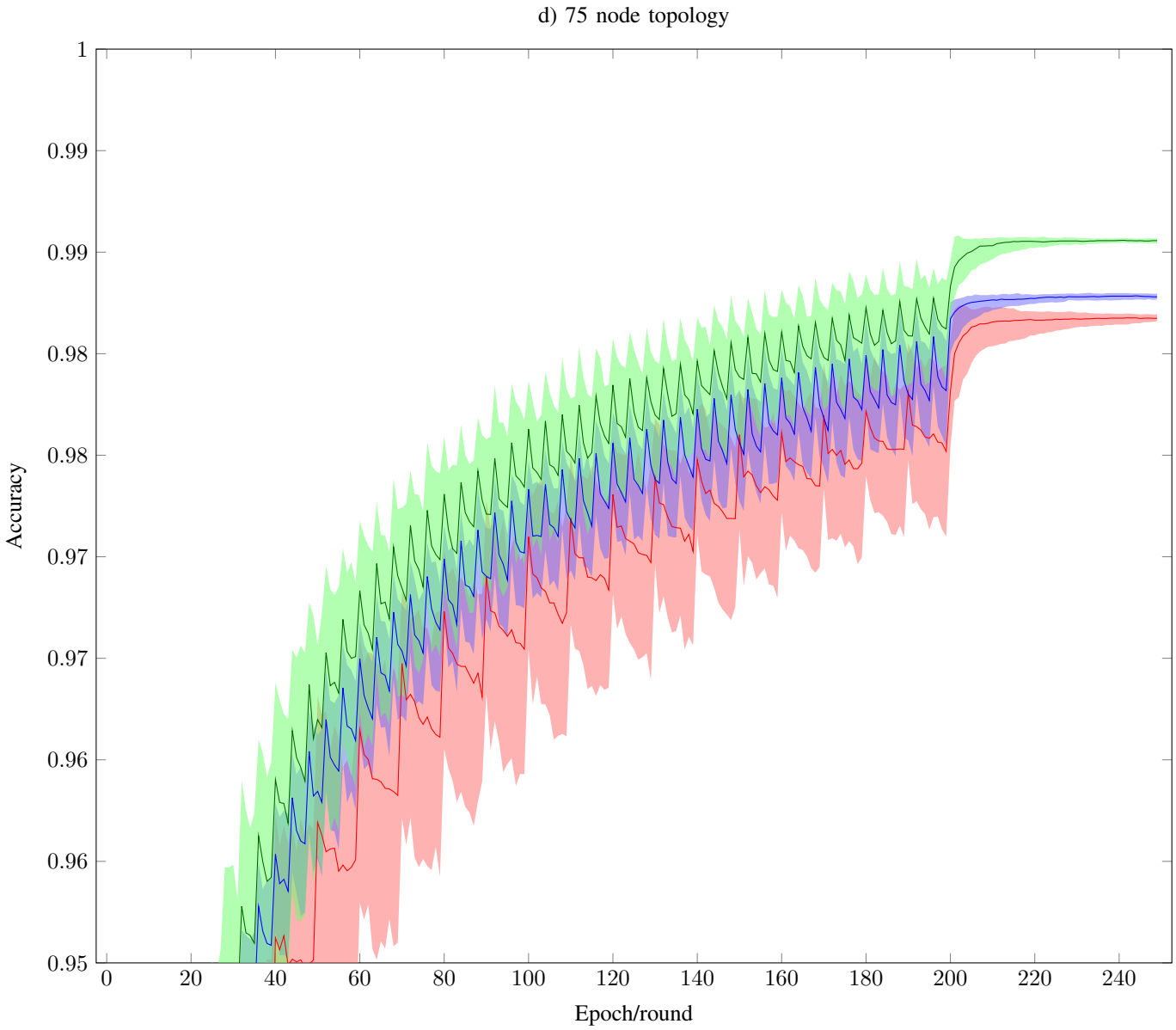



Fig. 31. Effect of increased update frequency $f \sim \sqrt{n}$ and full message propagation on both Delta Sum Learning and Averaging, compared to baseline Delta Sum Learning.

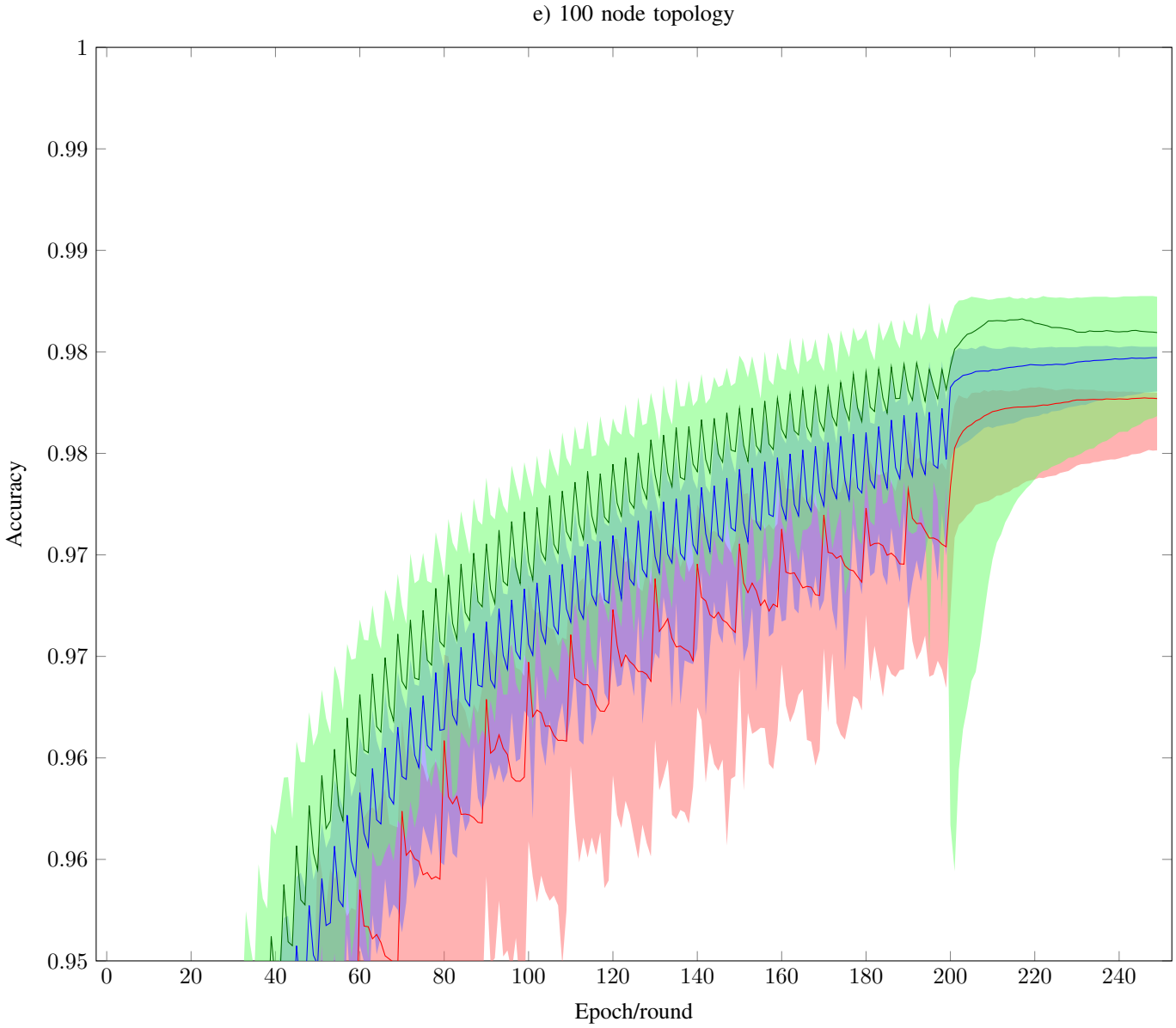



Fig. 32. Effect of increased update frequency $f \sim \sqrt{n}$ and full message propagation on both Delta Sum Learning and Averaging, compared to baseline Delta Sum Learning.

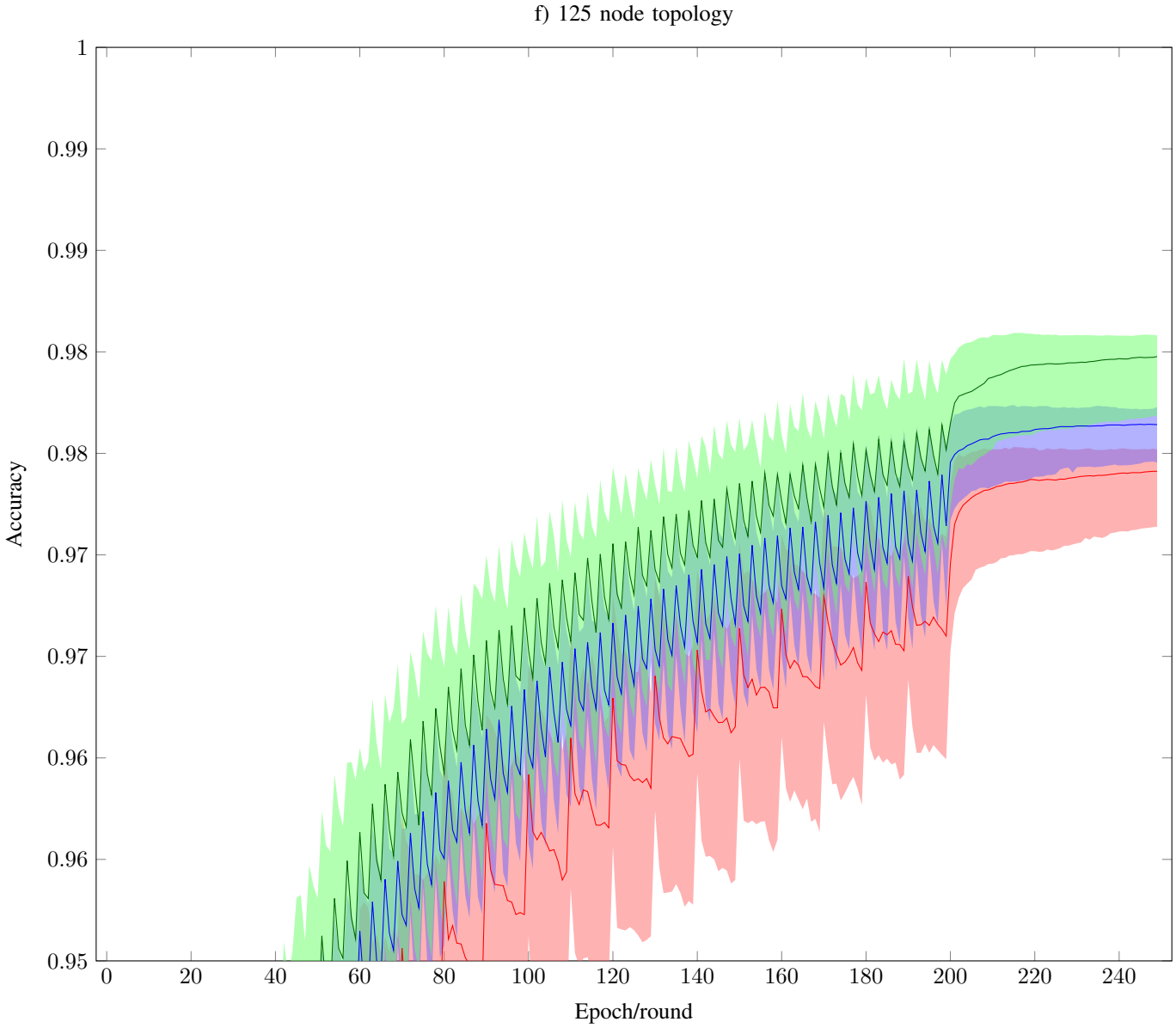



Fig. 33. Effect of increased update frequency $f \sim \sqrt{n}$ and full message propagation on both Delta Sum Learning and Averaging, compared to baseline Delta Sum Learning.

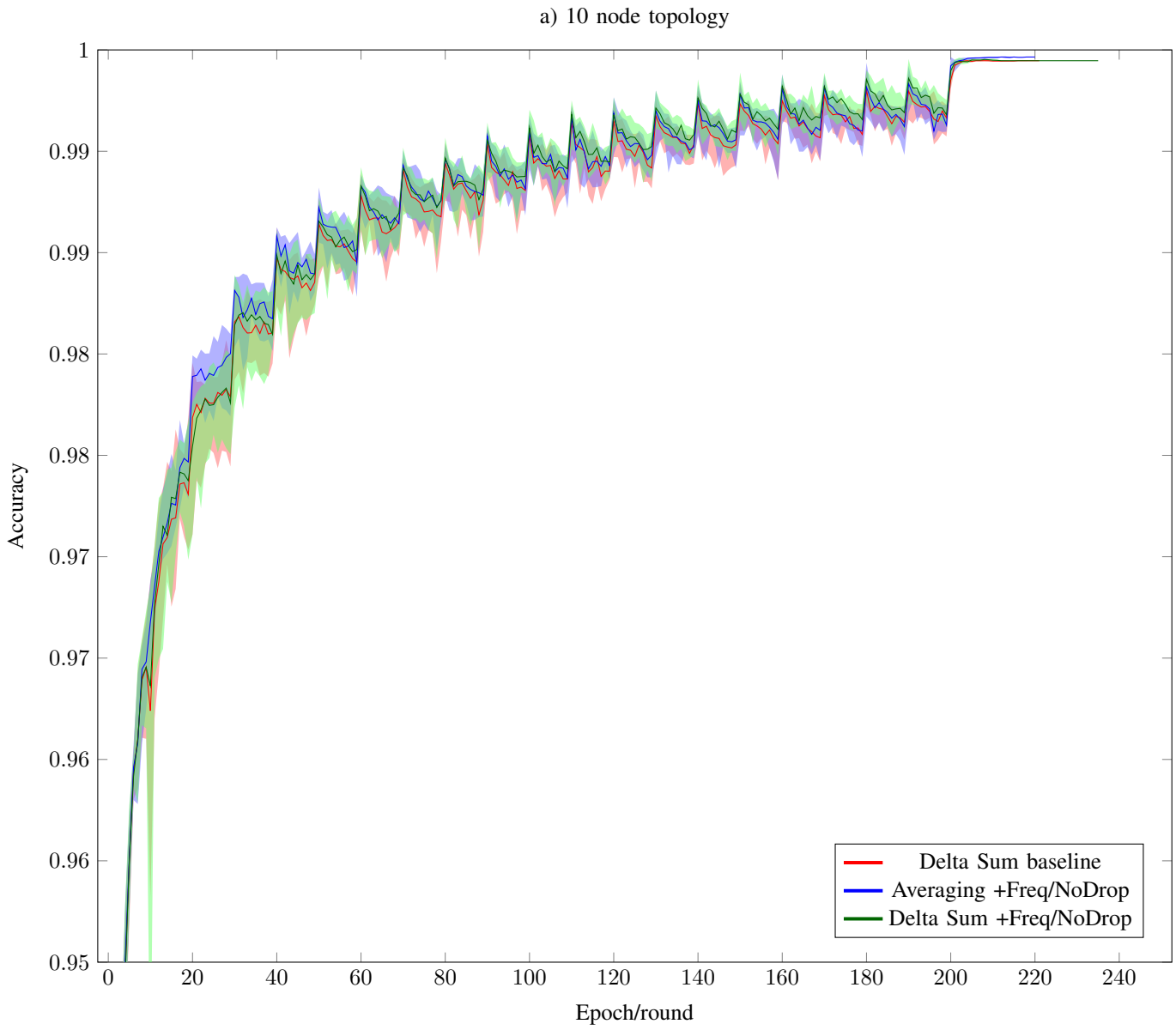



Fig. 34. Effect of variable connectivity and higher integration frequency/no gossip dropout on Delta Sum Learning and Averaging versus constant connectivity Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role.

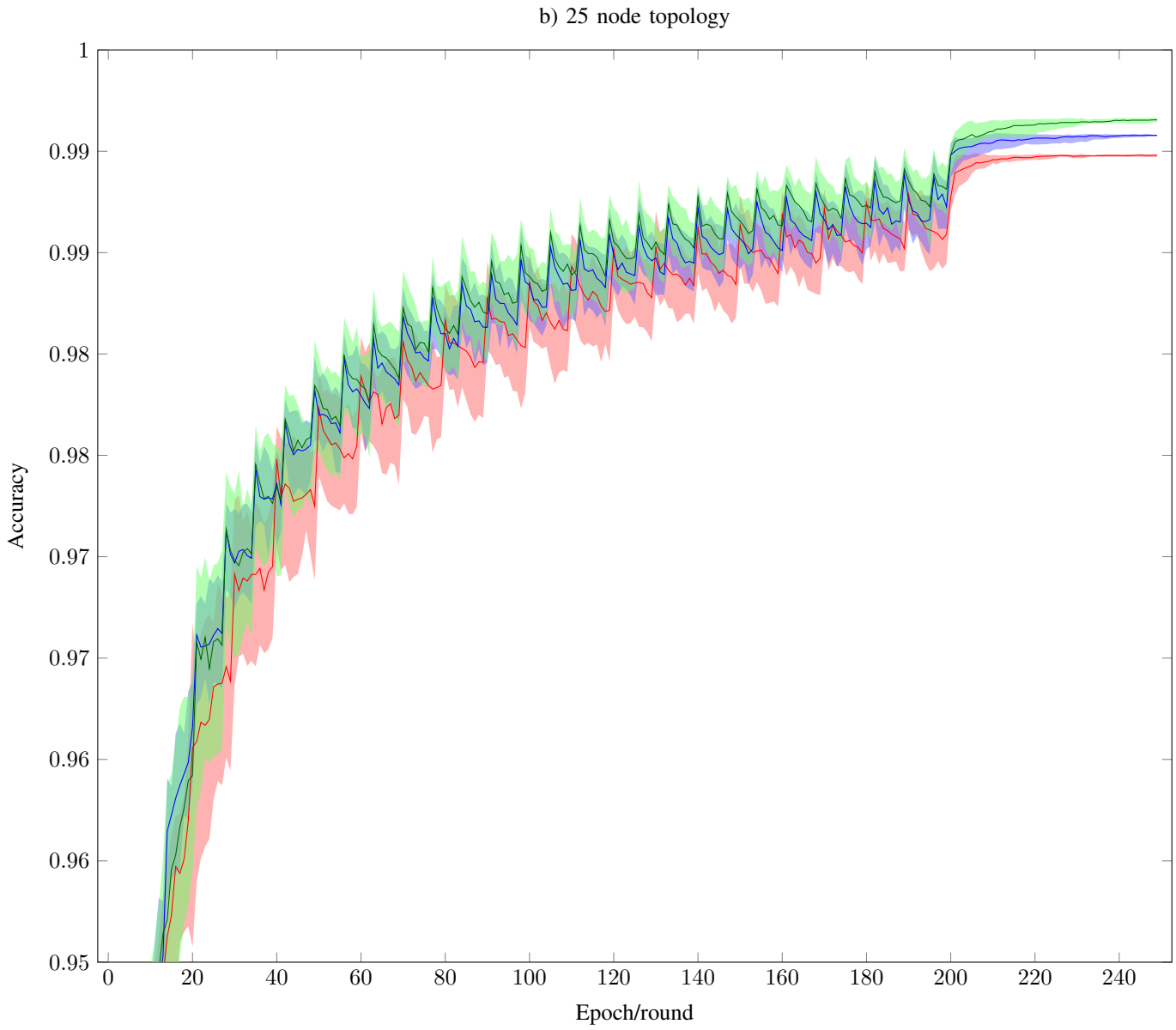



Fig. 35. Effect of variable connectivity and higher integration frequency/no gossip dropout on Delta Sum Learning and Averaging versus constant connectivity Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate “hub” role.

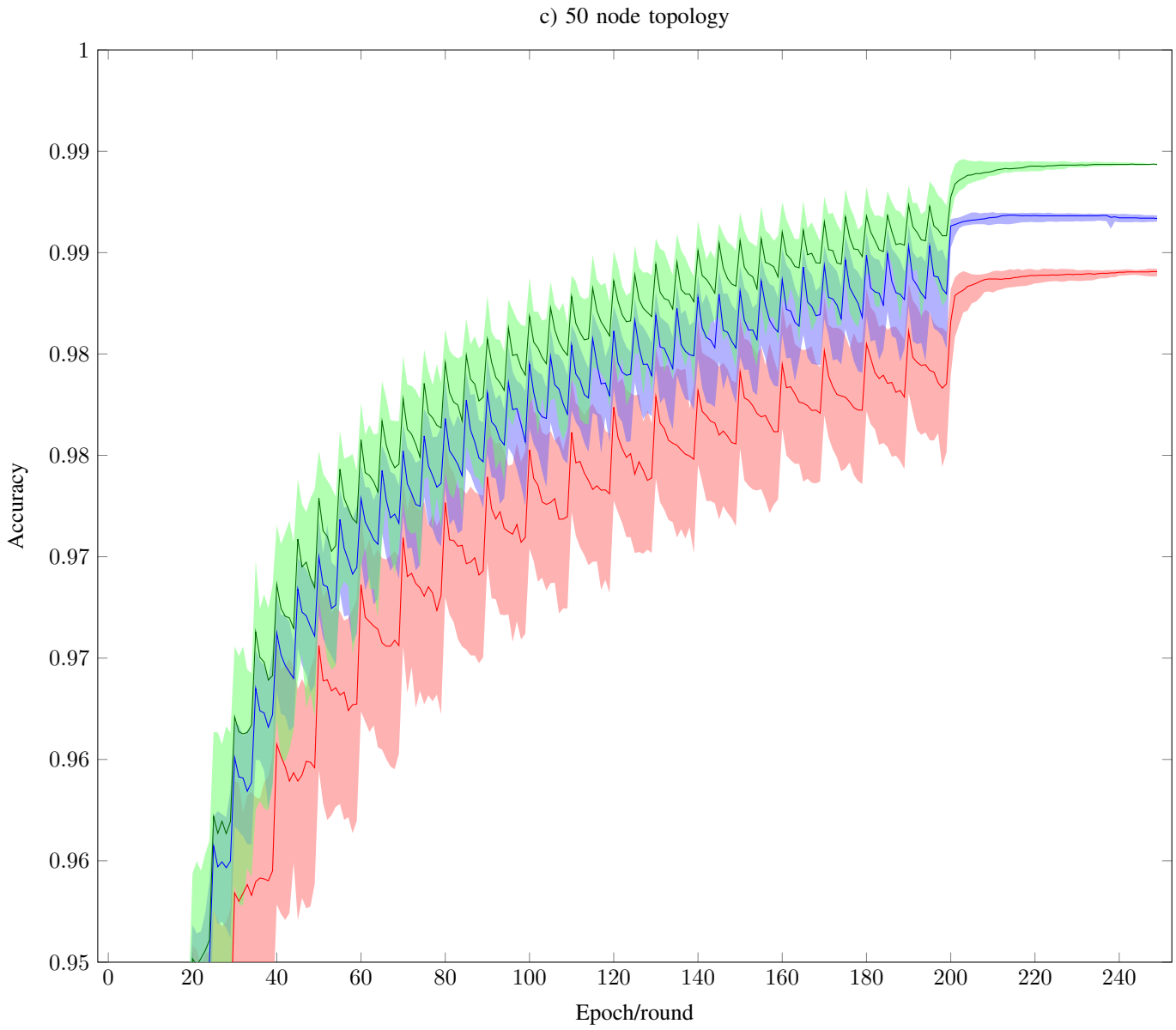



Fig. 36. Effect of variable connectivity and higher integration frequency/no gossip dropout on Delta Sum Learning and Averaging versus constant connectivity Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role.

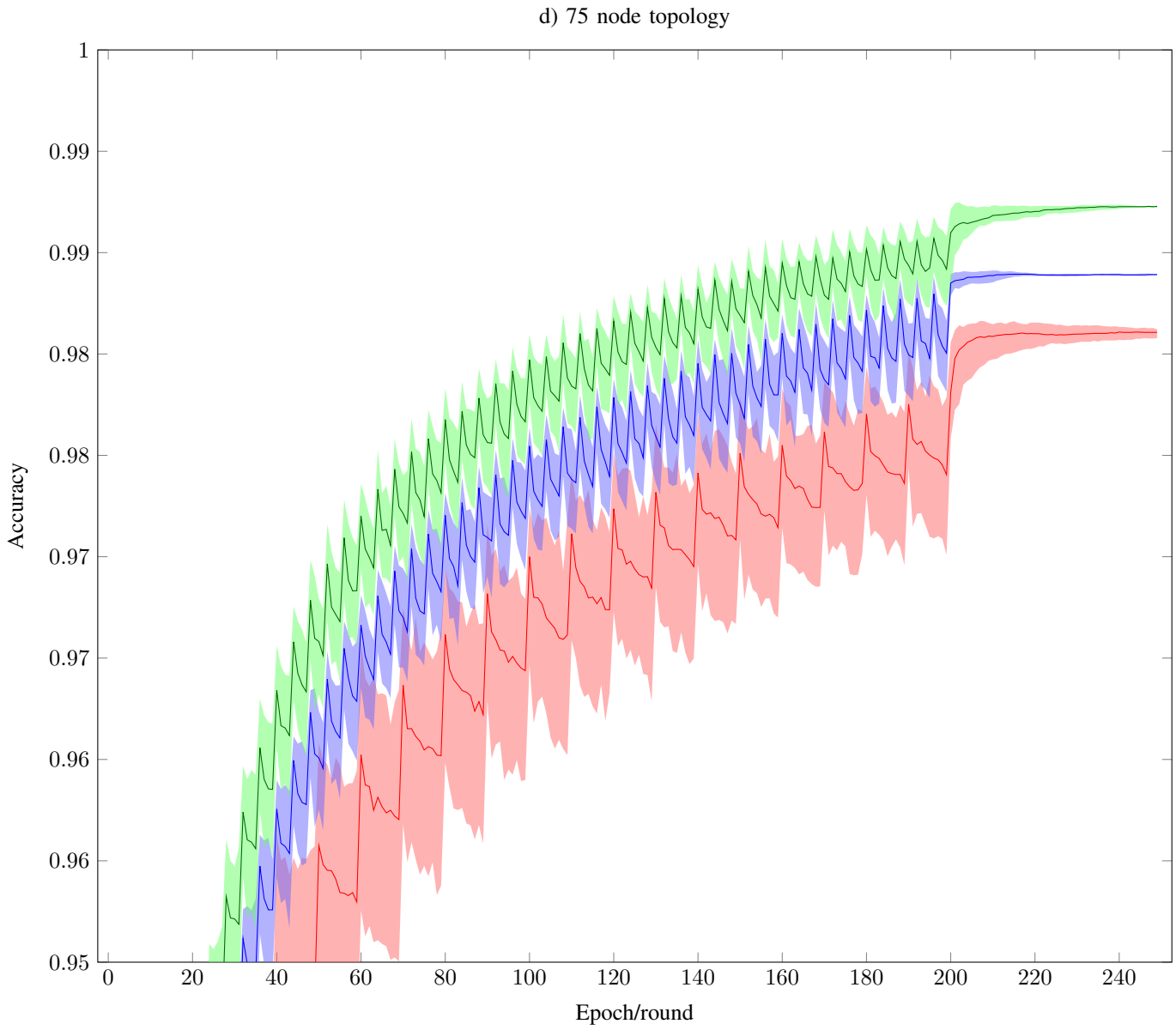



Fig. 37. Effect of variable connectivity and higher integration frequency/no gossip dropout on Delta Sum Learning and Averaging versus constant connectivity Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role.

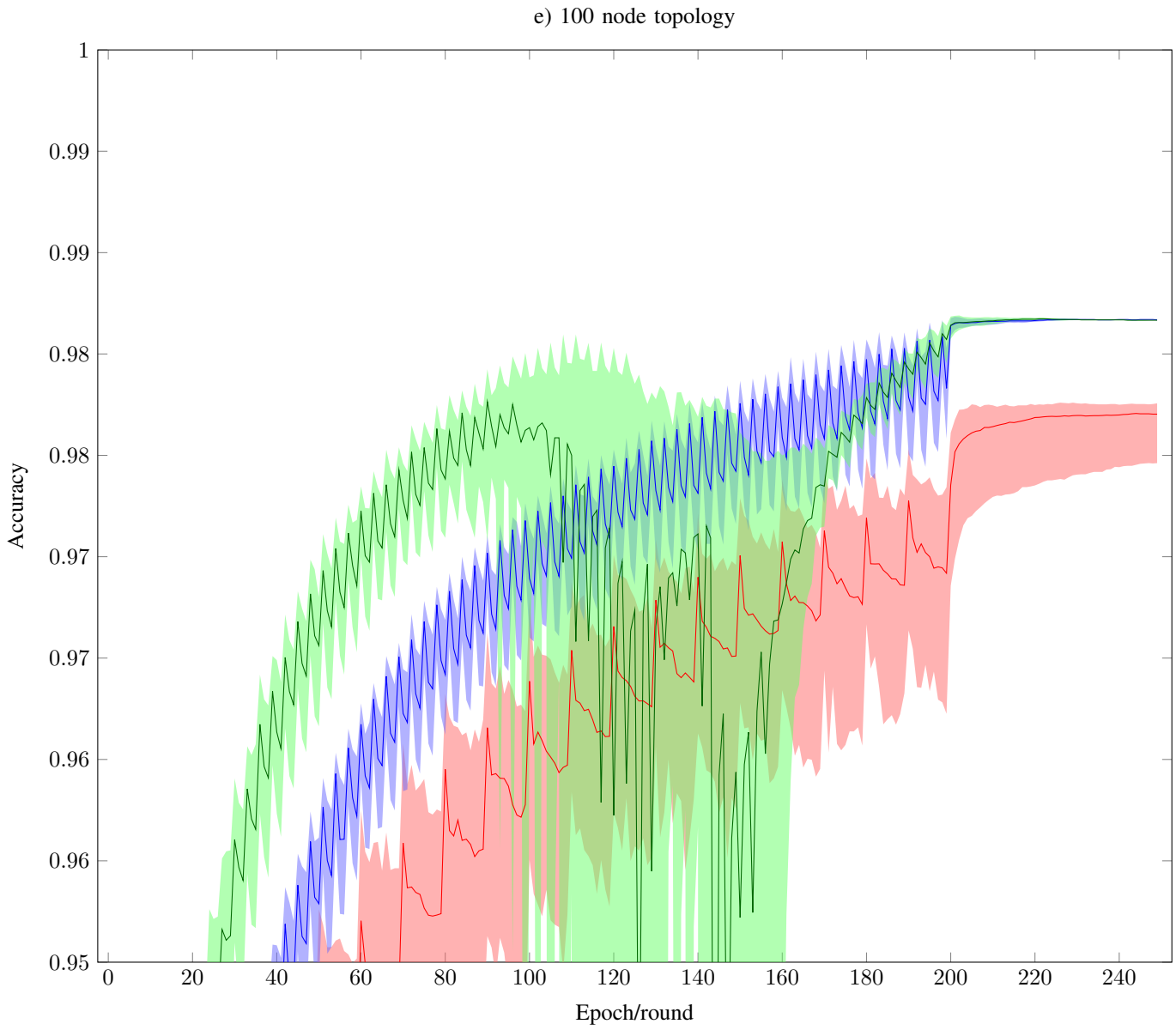



Fig. 38. Effect of variable connectivity and higher integration frequency/no gossip dropout on Delta Sum Learning and Averaging versus constant connectivity Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate "hub" role.

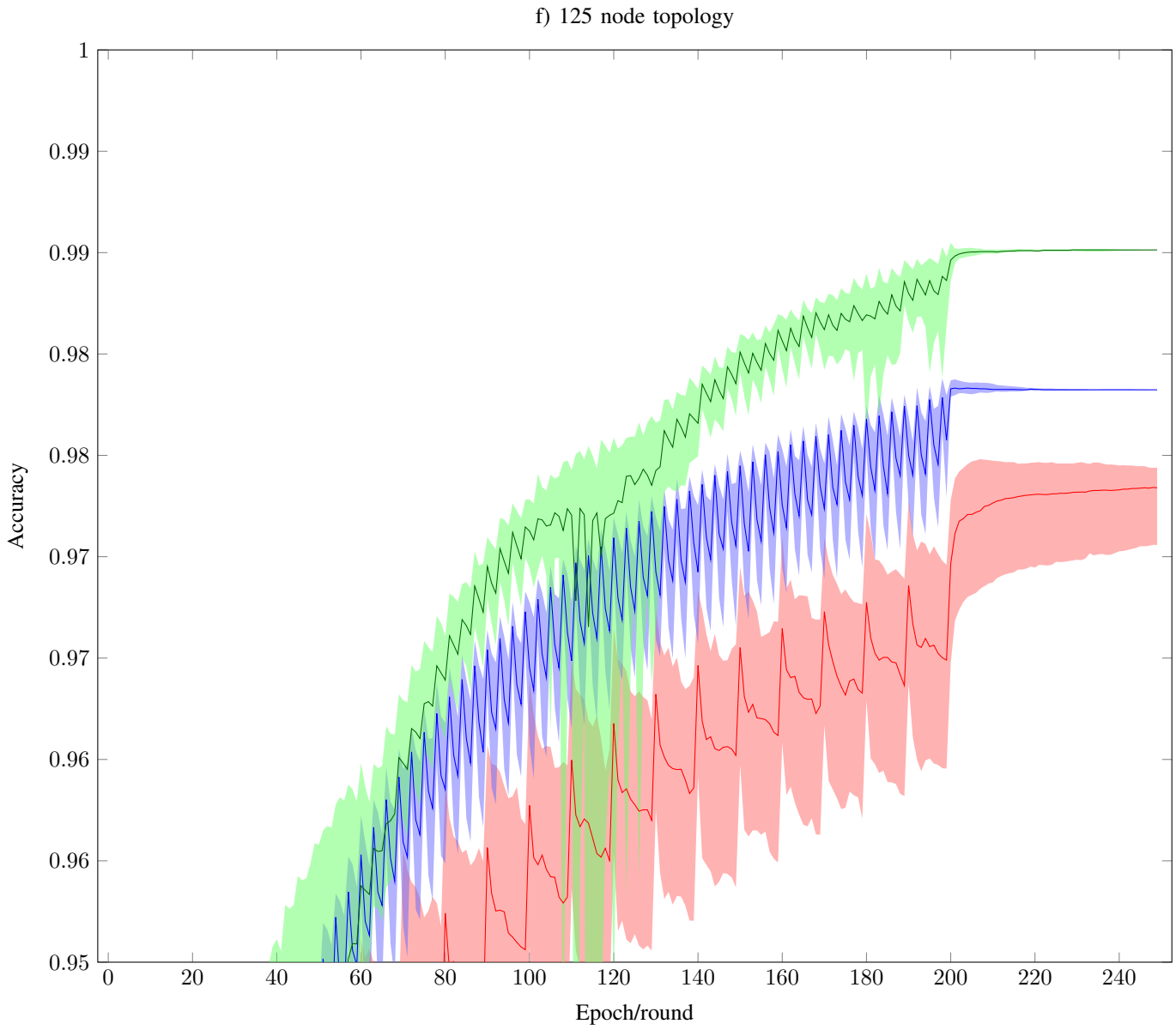



Fig. 39. Effect of variable connectivity and higher integration frequency/no gossip dropout on Delta Sum Learning and Averaging versus constant connectivity Delta Sum Learning. In topologies with variable connectivity, the number of possible neighbors for any specific node increases with total node count and varies by a factor of up to 50, giving some nodes a disproportionate “hub” role.

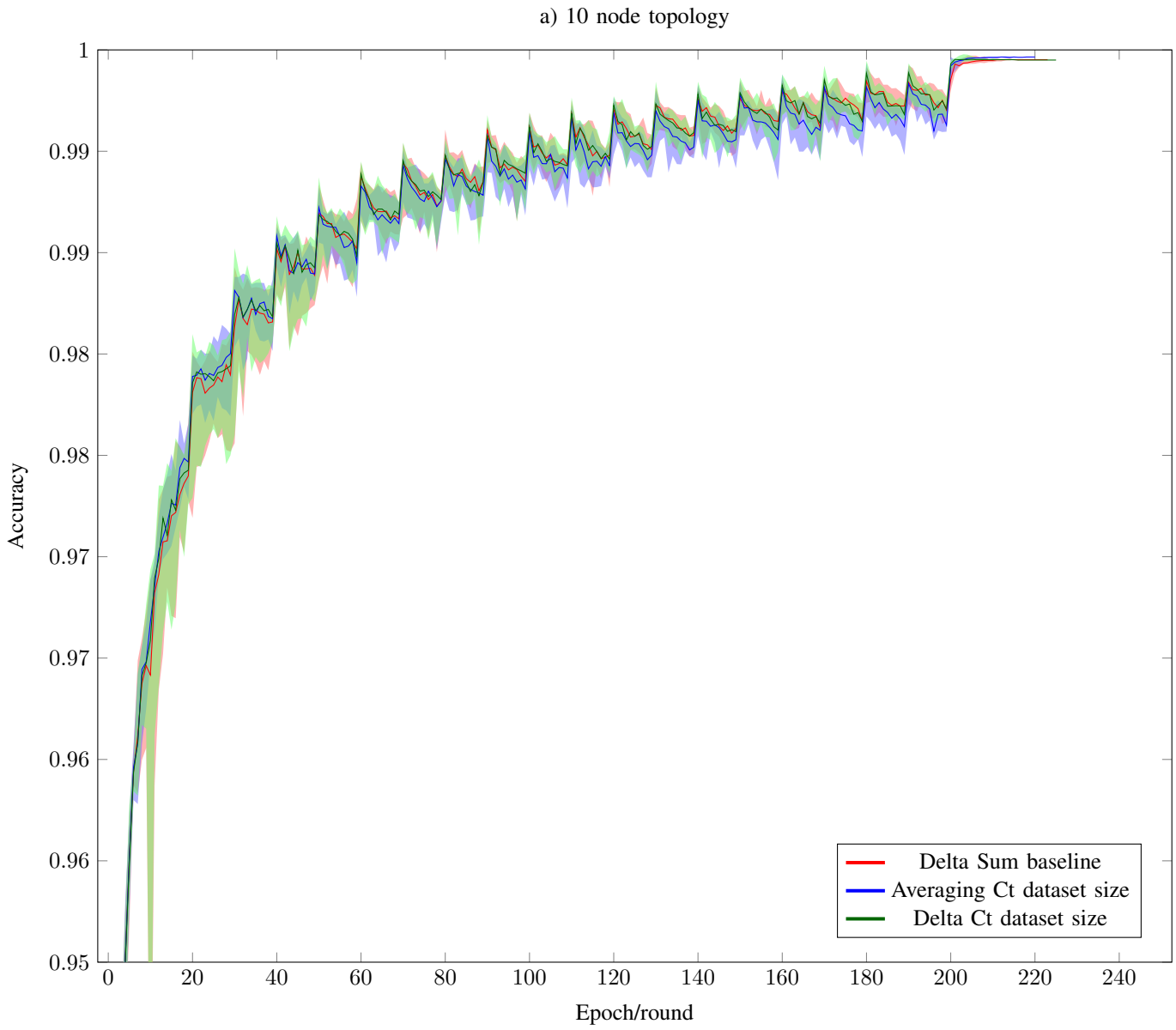



Fig. 40. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout. While these conditions are ideal for both Delta Sum Learning and Averaging, Delta Sum Learning, the latter outperforms Averaging by a relevant margin, especially in larger topologies.

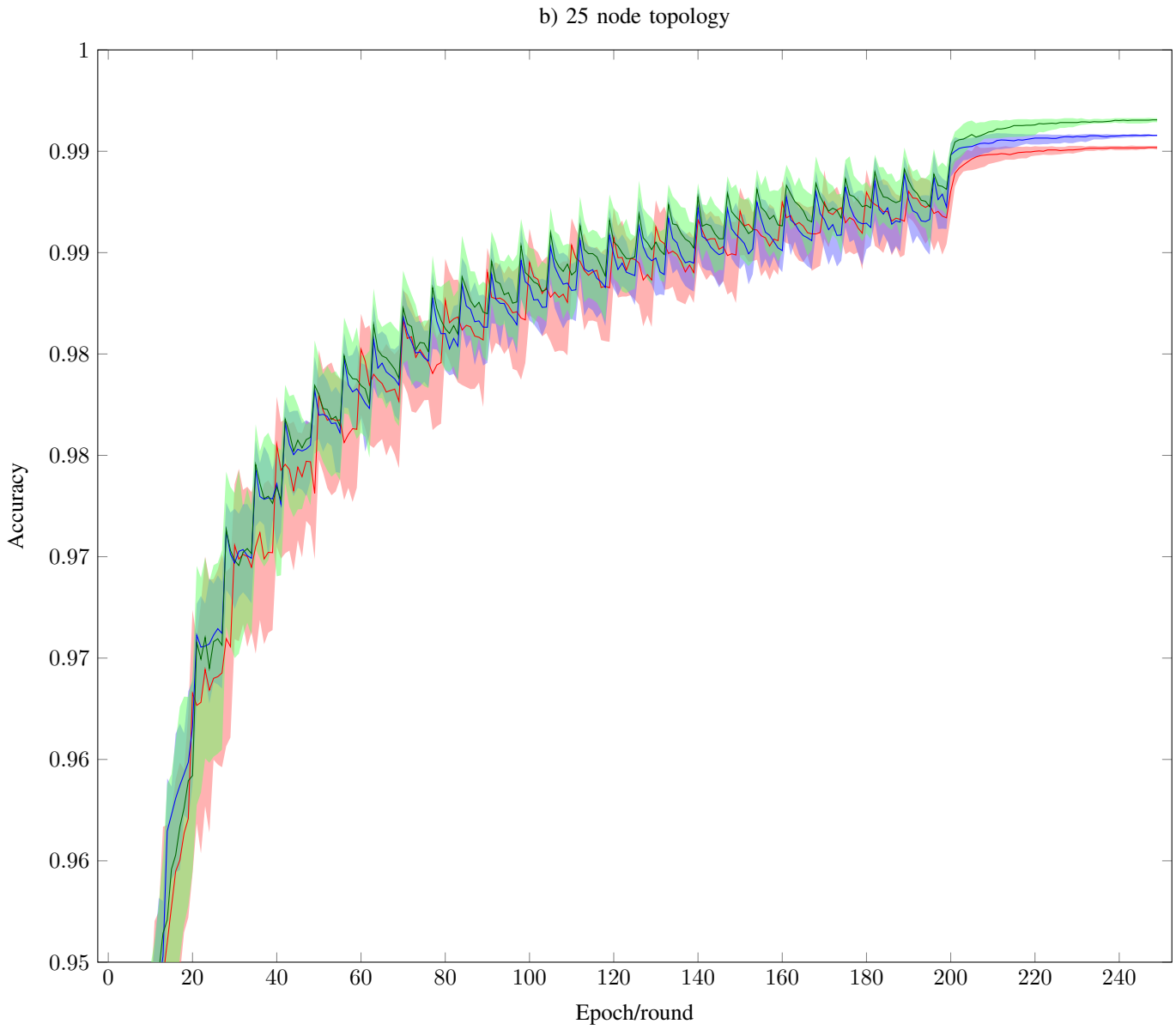



Fig. 41. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout. While these conditions are ideal for both Delta Sum Learning and Averaging, Delta Sum Learning, the latter outperforms Averaging by a relevant margin, especially in larger topologies.

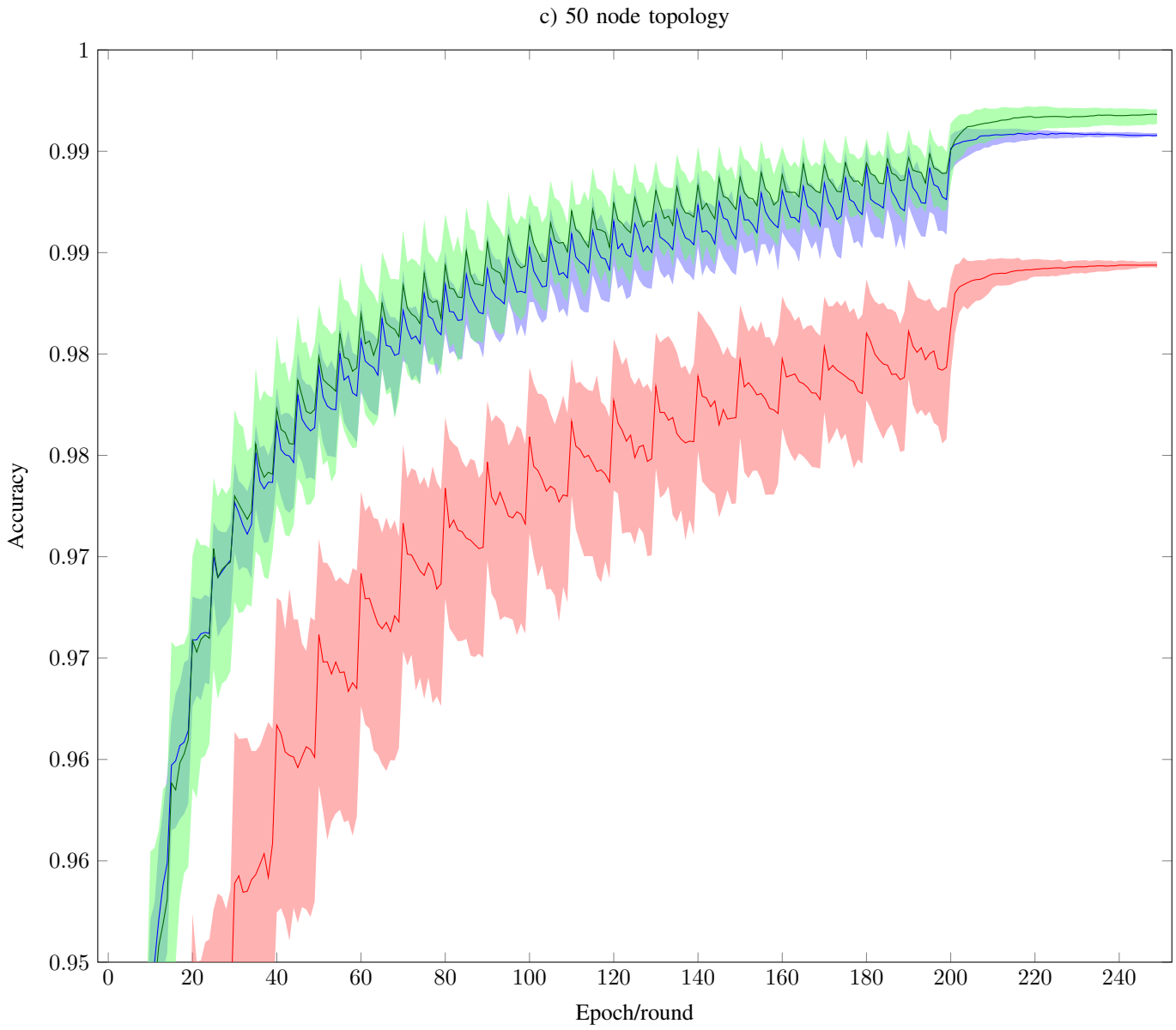



Fig. 42. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout. While these conditions are ideal for both Delta Sum Learning and Averaging, Delta Sum Learning, the latter outperforms Averaging by a relevant margin, especially in larger topologies.

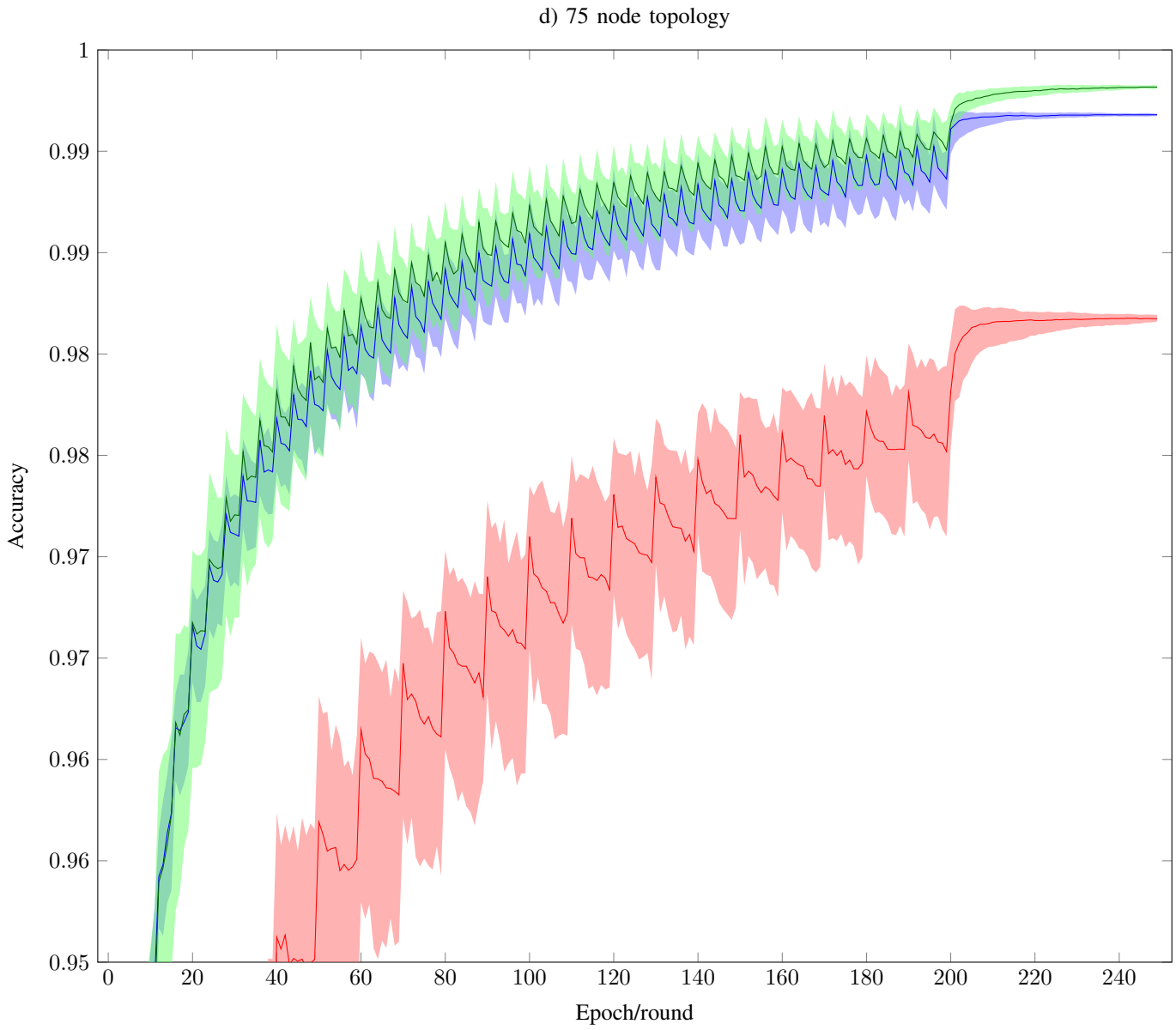



Fig. 43. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout. While these conditions are ideal for both Delta Sum Learning and Averaging, Delta Sum Learning, the latter outperforms Averaging by a relevant margin, especially in larger topologies.

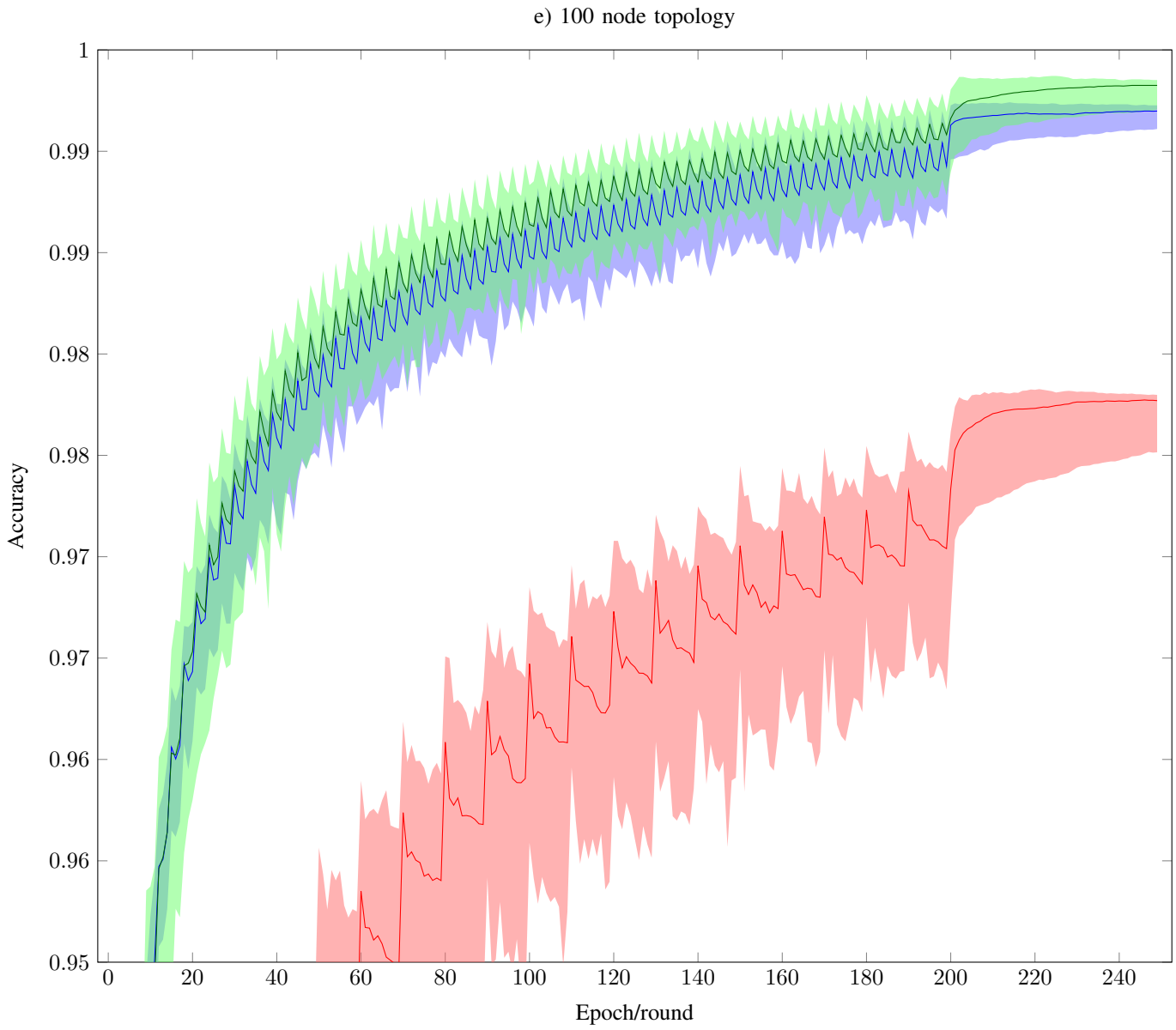



Fig. 44. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout. While these conditions are ideal for both Delta Sum Learning and Averaging, Delta Sum Learning, the latter outperforms Averaging by a relevant margin, especially in larger topologies.

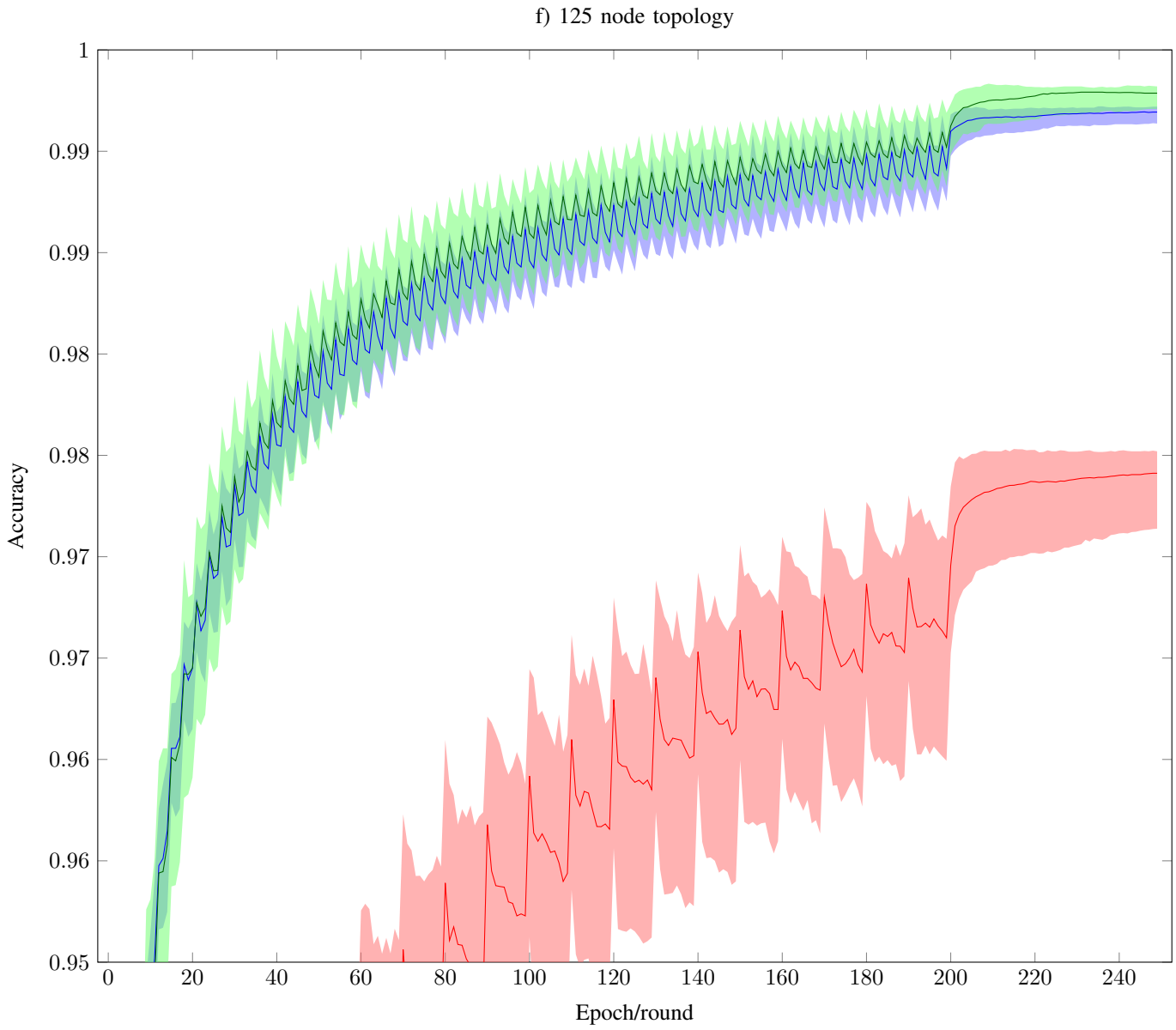



Fig. 45. The effects of diminishing dataset size and increasing node training dataset variance (baseline) vs. constant dataset size per node (normalized to 25 nodes), and the additional effects of integration frequency and disabling gossip message dropout. While these conditions are ideal for both Delta Sum Learning and Averaging, Delta Sum Learning, the latter outperforms Averaging by a relevant margin, especially in larger topologies.